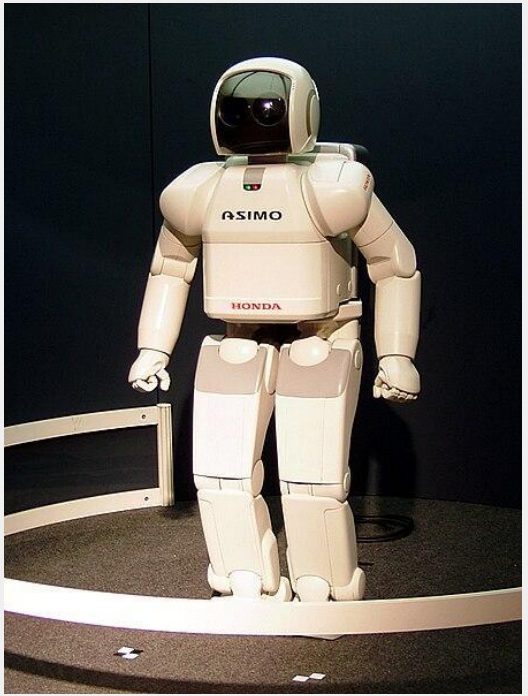
History of AI

Global Perspectives in History of AI



unsojo@cornell.edu

Lecture 3 Februrary 7, 2025

Hawley Introduces Legislation to Decouple American AI Development from Communist China

Wednesday, January 29, 2025

Today, U.S. Senator Josh Hawley (R-Mo.) introduced legislation to protect America's artificial intelligence (AI) development from China.

"Every dollar and gig of data that flows into Chinese AI are dollars and data that will ultimately be used against the United States," **said Senator Hawley.** "America cannot afford to empower our greatest adversary at the expense of our own strength. Ensuring American economic superiority means cutting China off from American ingenuity and halting the subsidization of CCP innovation."

Senator Hawley's new legislation comes after China unveiled "DeepSeek," a data-harvesting, low-cost AI model that sparked international concern and sent American technology stocks plummeting.

Senator Hawley's *Decoupling America's Artificial Intelligence Capabilities from China Act* would:

- Prohibit the import from or export to China of artificial intelligence technology;
- Prohibit American companies from conducting AI research in China or in cooperation with Chinese companies; and
- Prohibit U.S. companies from investing money in Chinese AI development.

Sam Altman's Answer to DeepSeek Is Giving Away OpenAI's Tech

CEO of the ChatGPT maker says his company has been 'on the wrong side of history' with open-source software

By Deepa Seetharaman Follow
Jan. 31, 2025 7:50 pm ET

Share Resize Listen (1 min)



OpenAI CEO Sam Altman. PHOTO: ANDREW HARNIK/GETTY IMAGES

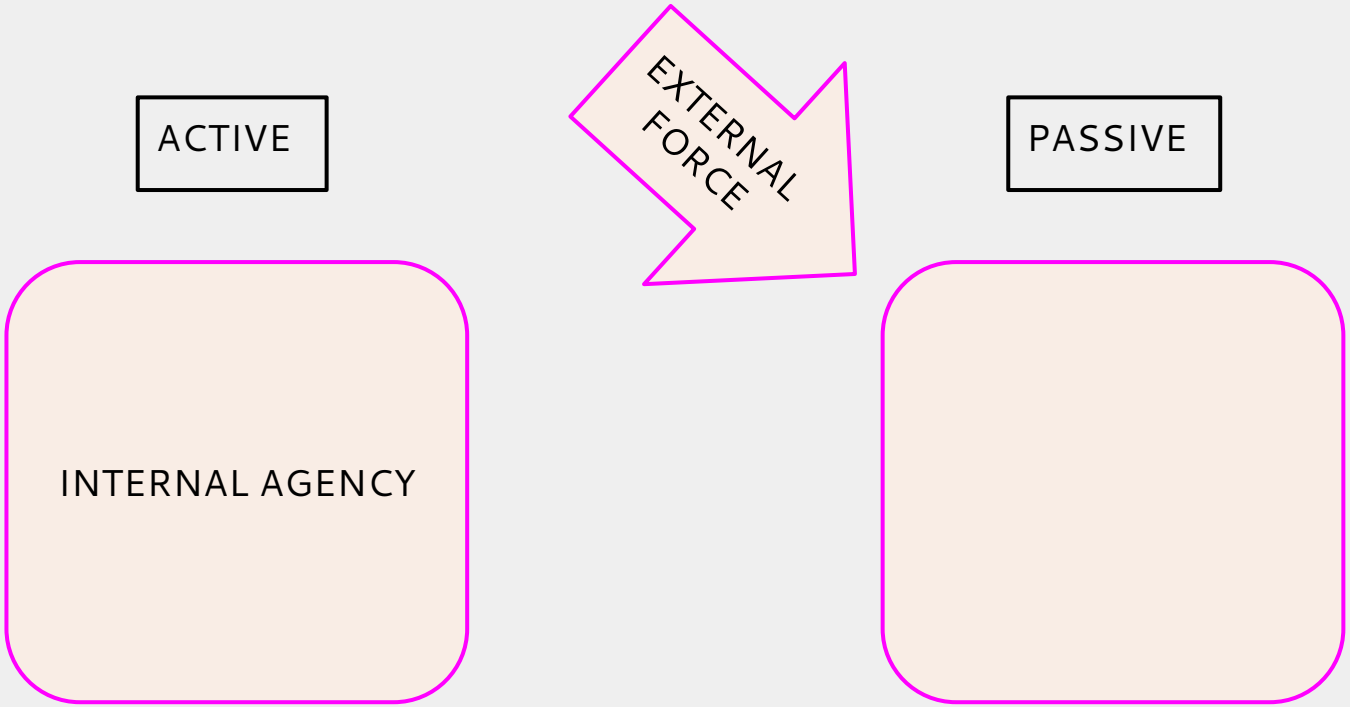
OpenAI Chief Executive Sam Altman said he believes his company should consider giving away its AI models, a potentially seismic strategy shift in the same week China's DeepSeek has upended the artificial-intelligence industry.

Last Week!

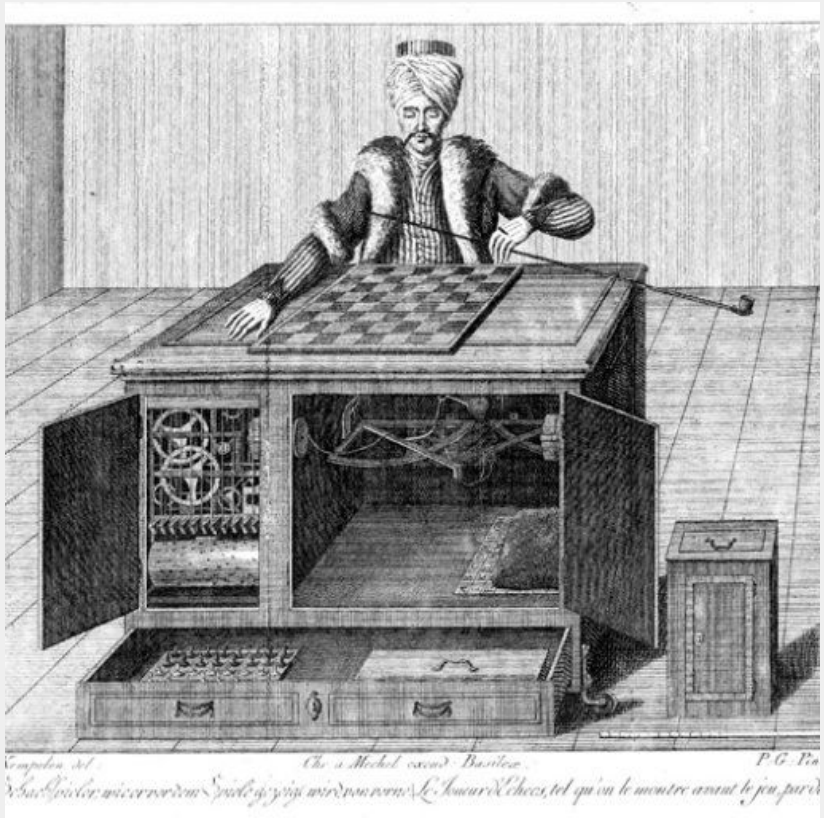
- New Science
- Faith, AI, & Agency
- Degrees of Autonomousness:
  - Able to repeat tasks
  - Able to take varying input
  - Able to select input
  - Able to learn from experience

Early Modern

17thC Debates on “Agency” in Nature



Man-Machines



Chess-Playing Turk Wolfgang von Kempelen 1782

Punch Cards

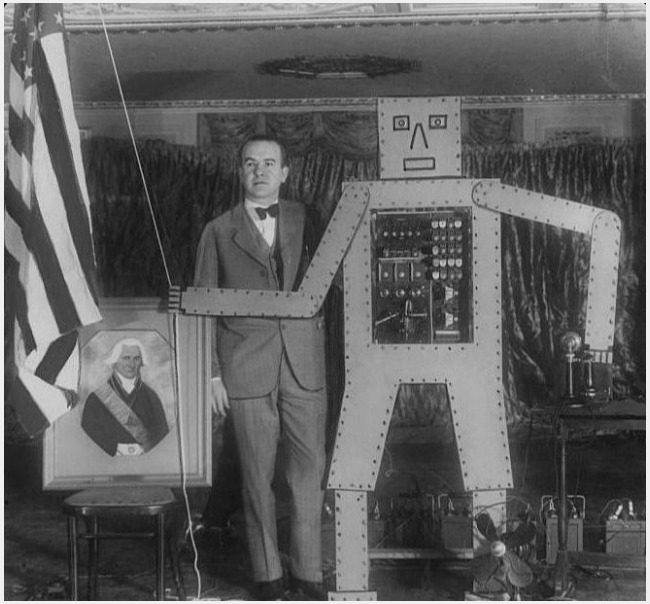


Jacquard Loom 1804



Jacquard woven on Silk. 1839

Man-Machines of 20th C



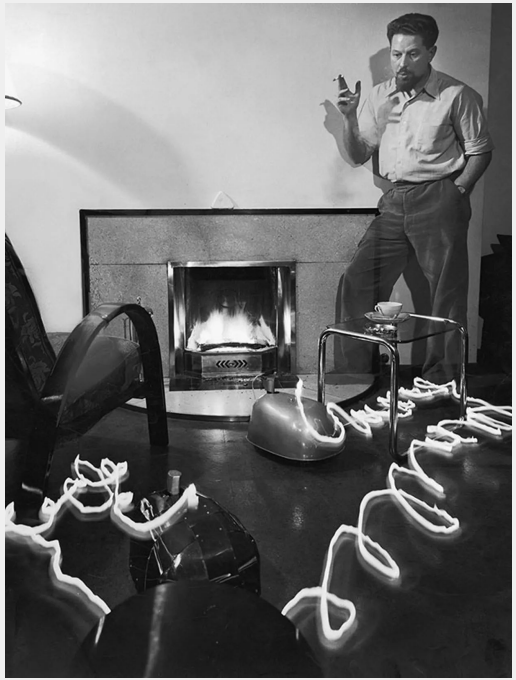
Mr. Televox. R.J. Wensley, Westinghouse (Westinghouse Electrical Company) 1926

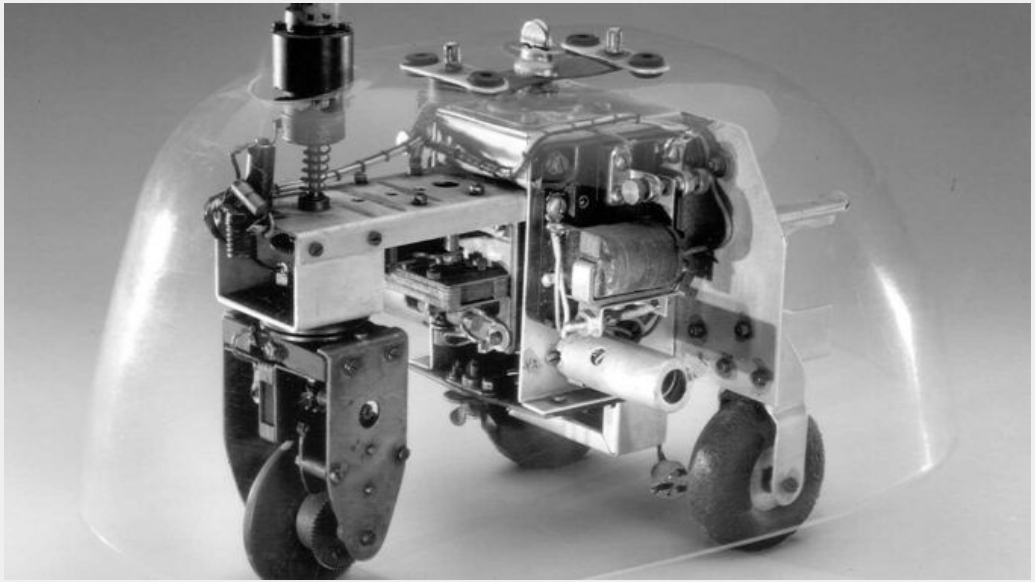


Eric the robot at Model Engineer Exhibition at the Royal Agricultural Hall in London. 1928

Cybernetics

Sentience in terms of their outward manifestations not internal mechanism or subjective experience



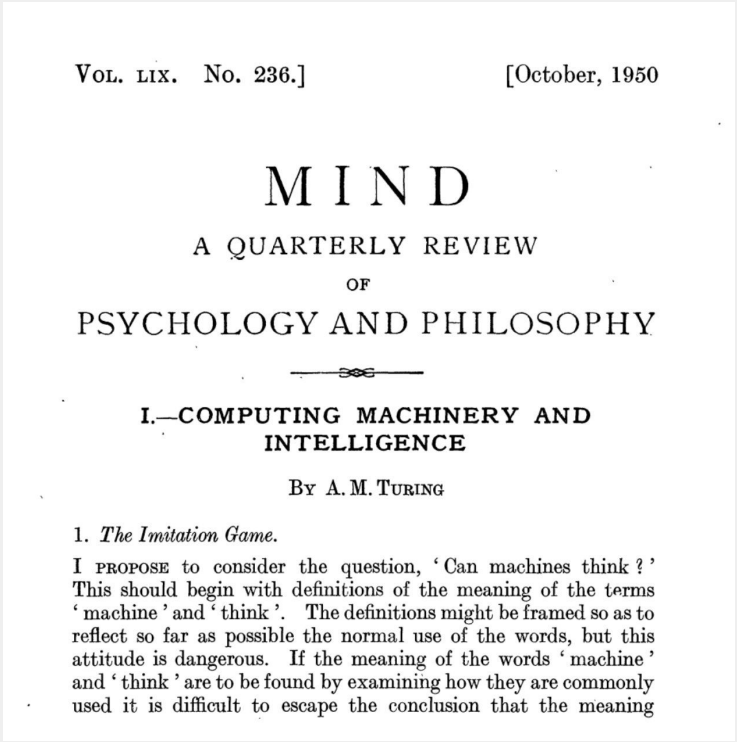


Elmer and Elsie 1948 (Grey Walter)

Turing

Intelligence develops through learning

- 1950 “Computers Machinery and Intelligence”:
- Pleasure/Reward vs. Pain/Punishment
- “Machines must be fallible”
- Mathematical equations can’t determine whether a machine can think



VOL. LIX. No. 236.] [October, 1950

MIND

A QUARTERLY REVIEW

OF

PSYCHOLOGY AND PHILOSOPHY

I.—COMPUTING MACHINERY AND INTELLIGENCE

BY A. M. TURING

1. *The Imitation Game.*

I PROPOSE to consider the question, ‘Can machines think?’ This should begin with definitions of the meaning of the terms ‘machine’ and ‘think’. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words ‘machine’ and ‘think’ are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning

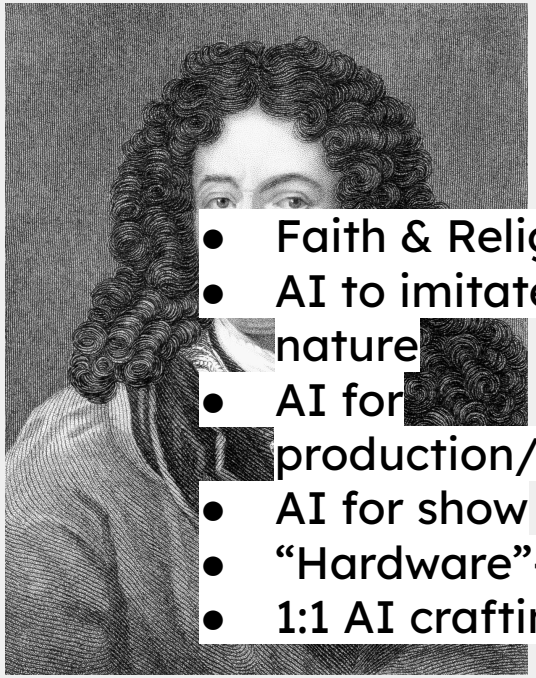
Early Modern



Modern



Early Modern



- Faith & Religion
- AI to imitate God/mimic nature
- AI for production/productivity
- AI for show
- “Hardware”-focused
- 1:1 AI crafting

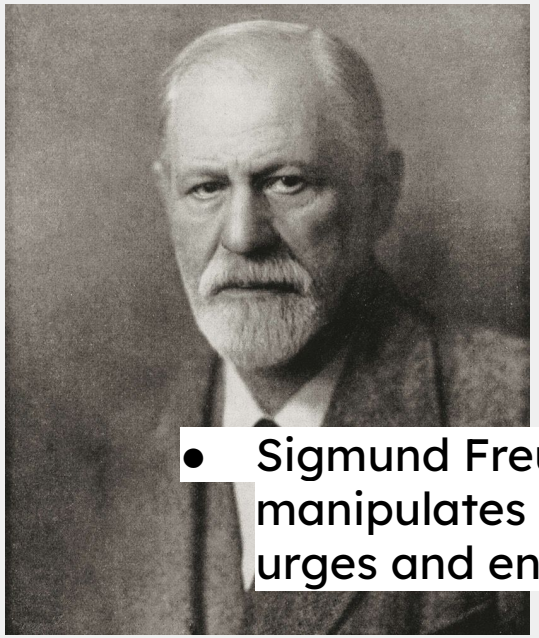
Modern



- Institutions (Politics, Nation States)
- AI for destruction
- AI with functionality
- Imitating the mind/algorithms
- Increased interactivity
- scaled-production

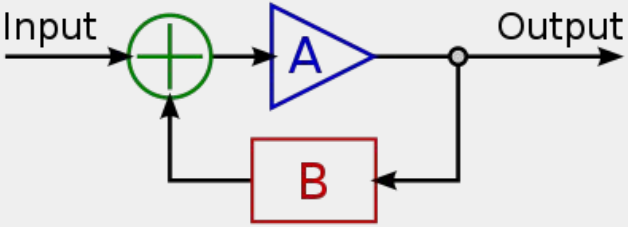
- What does it mean to “modernize” a scientific field?
- The role of defense in the origins of modern AI. How is AI defense changing warfare?
- Military Industrial Complex – is “invention the mother of necessity”
- Which came first: AI or computers?

Early Modern



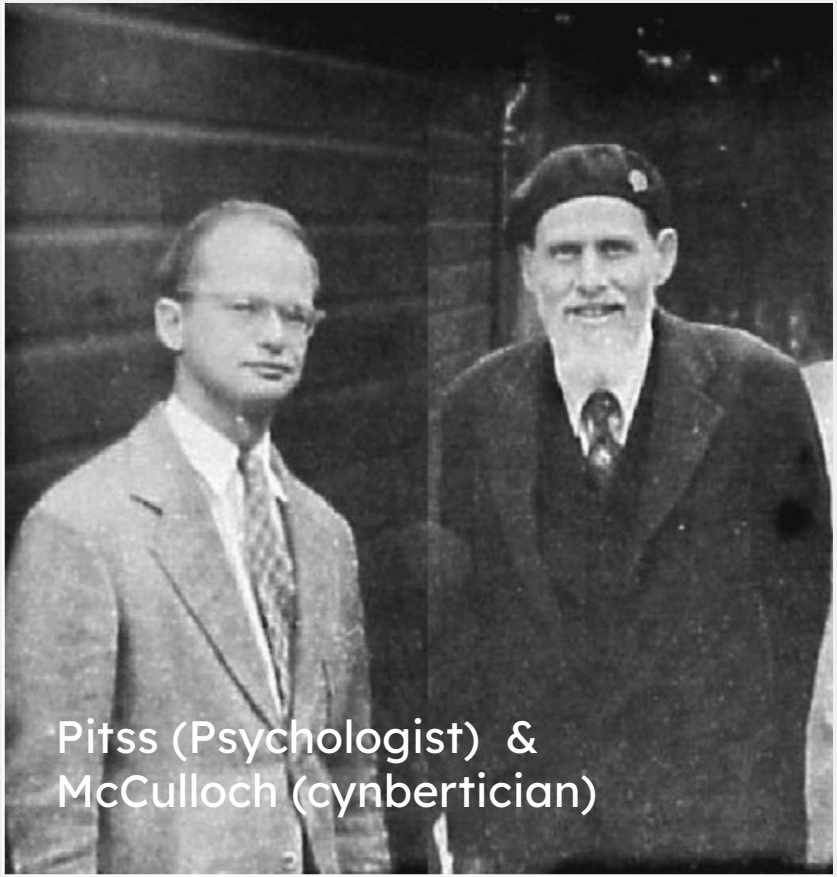
- Sigmund Freud: Mind manipulates biological urges and energies

Modern



- Intelligence is about taking, processing information

Pitts-McCulloch 1943



A LOGICAL CALCULUS OF THE IDEAS IMMANENT IN NERVOUS ACTIVITY

WARREN S. MCCULLOCH and WALTER H. PITTS

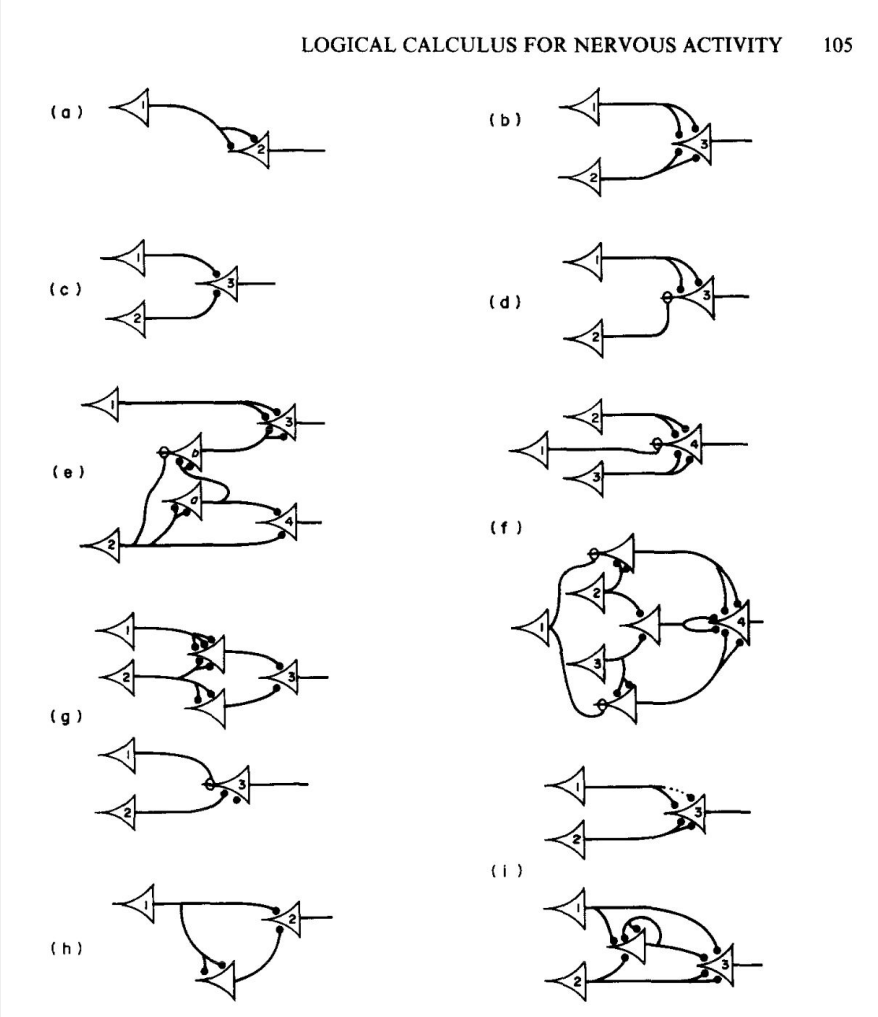
Because of the "all-or-none" character of nervous activity, neural events and the relations among them can be treated by means of propositional logic. It is found that the behavior of every net can be described in these terms, with the addition of more complicated logical means for nets containing circles; and that for any logical expression satisfying certain conditions, one can find a net behaving in the fashion it describes. It is shown that many particular choices among possible neurophysiological assumptions are equivalent, in the sense that for every net behaving under one assumption, there exists another net which behaves under the other and gives the same results, although perhaps not in the same time. Various applications of the calculus are discussed.

INTRODUCTION

THEORETICAL neurophysiology rests on certain cardinal assumptions. The nervous system is a net of neurons, each having a soma and an axon. Their adjunctions, or synapses, are always between the axon of one neuron and the soma of another. At any instant a neuron has some threshold, which excitation must exceed to initiate an impulse. This, except for the fact and the time of its occurrence, is determined by the neuron, not by the excitation.

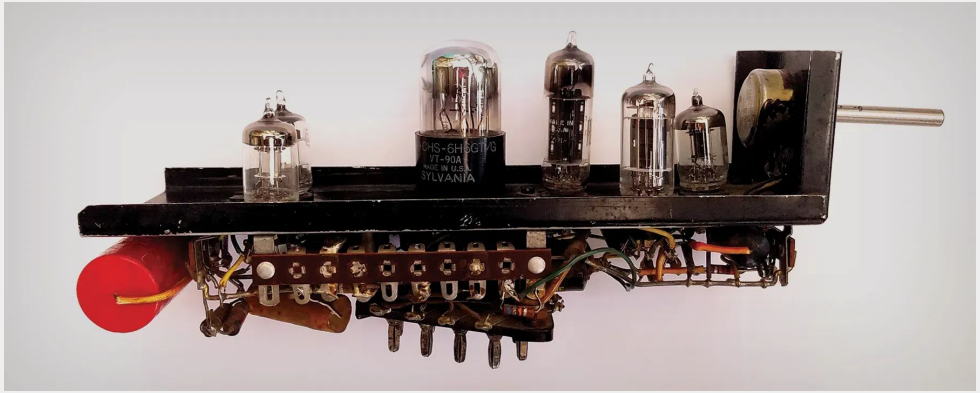
Pitts-McCulloch 1943

- inspired by feedback loops
- “can represent any logical expression with neural nets”
- “better than Turing machine!”



Marvin Minsky's SNARC 1951

Pitts-McCulloch neural nets + B.F. Skinner reinforcement learning



1 neuron!

- made out of b-24 bomber aircraft pieces
- 40 neurons
- 6 vacuum tubes each
- size of small room
- was able to solve Shannon's maze
- → sufficient memory for reasoning would require millions of neurons

Dartmouth Conference - an interdisciplinary AI conference 1957

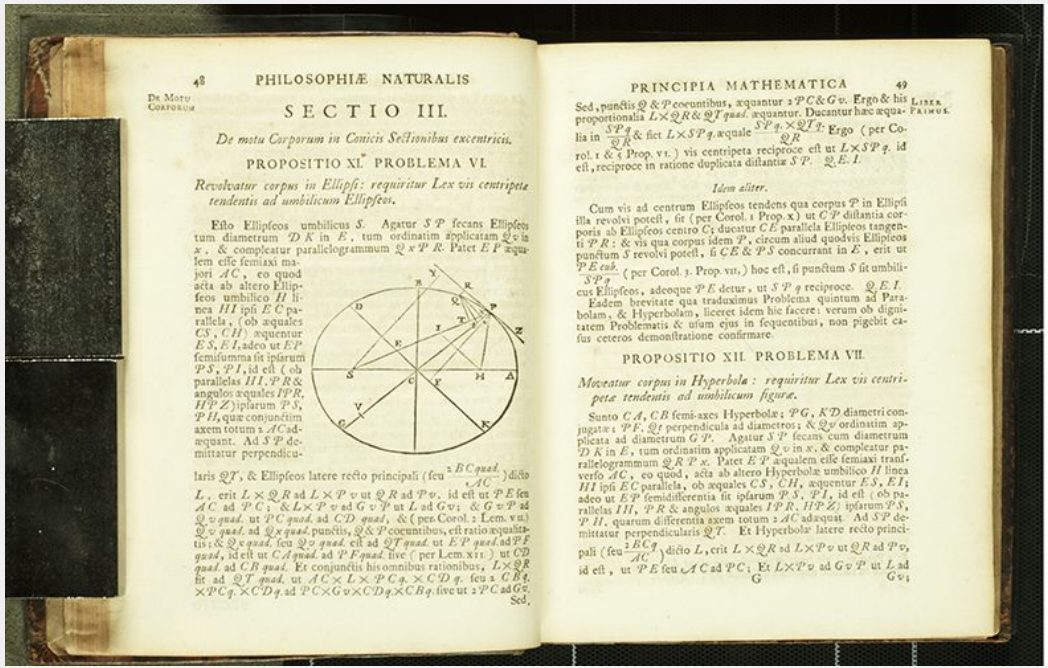
"We propose that a 2 month, 10 man (**John McCarthy hosted; Minsky, Claude Shannon, Arthur Samuel, Ray Solomonoff, etc.**) study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hamsphire."

→ Stanford, CMU, MIT, IBM



Dartmouth Conference 1957

First? AI Program - Logic Theorist 1955 (Newell, Simon, Shaw)



48 PHILOSOPHIÆ NATURALIS

De Motu Corporum

SECTIO III.

De motu Corporum in Conicis Sectionibus excentricis.

PROPOSITIO XI. PROBLEMA VI.

Revolvatur corpus in Ellipſi: requiritur Lex vis centripetæ tendentis ad umbilicum Ellipſeos.

Eſto Ellipſeos umbilicus *S*. Agatur *SP* fecans Ellipſeos tum diametrum *DK* in *E*, tum ordinatim applicatam *Qv* in *x*, & compleatur parallelogrammum *QxPR*. Patet *EP* æqualem eſſe ſemiaxi majori *AC*, eo quod acta ab altero Ellipſeos umbilico *H* linea *HI* ipſi *EC* parallela, (ob æquales *CS*, *CH*) æquentur *ES*, *EI*, adeo ut *EP* ſemiſumma ſit ipſarum *PS*, *PI*, id eſt, (ob parallelas *HI*, *PR* & angulos æquales *IPR*, *HPZ*) ipſarum *PS*, *PH*, quæ conjunctim axem totum 2*AC* adæquant. Ad *SP* demittatur perpendicularis *QT*, & Ellipſeos latere recto principali (ſeu $\frac{2BC\ quad.}{AC}$) dicto *L*, erit $L \times QR$ ad $L \times Pv$ ut *QR* ad *Pv*, id eſt ut *PE* ſeu *AC* ad *PC*; & $L \times Pv$ ad Gv ut *L* ad *Gv*; & *Gv* ad *Qv quad.* ut *PC quad.* ad *CD quad.*, & (per Corol. 2 Lem. VII.) *Qv quad.* ad *Qx quad.* punctis, *Q* & *P* coeuntibus, eſt ratio æqualitatis; & *Qx quad.* ſeu *Qv quad.* eſt ad *QT quad.* ut *EP quad.* ad *PF quad.*, id eſt ut *CA quad.* ad *PF quad.* hoc eſt (per Lem. XII.) ut *CD quad.* ad *CB quad.* Et conjunctis his omnibus rationibus, $L \times QR$ fit ad *QT quad.* ut $AC \times L \times PC q. \times CD q.$ ſeu 2*CB q.* $\times PC q. \times CD q.$ ad $PC \times Gv \times CD q. \times CB q.$ hoc eſt ut 2*PC* ad *Gv*. Sed,

PRINCIPIA MATHEMATICA 49

Sed, punctis *Q* & *P* coeuntibus, æquantur 2*PC* & *Gv*. Ergo & his proportionalia $L \times QR$ & *QT quad.* æquantur. Ducantur hæc æqualia in $\frac{SP}{QR}$, & fiet $L \times SP\ q.$ æquale $\frac{SP\ q. \times QT\ q.}{QR}$. Ergo (per Corol. 1 & 2 Prop. VI.) vis centripeta reciproce eſt ut $L \times SP\ q.$ id eſt, reciproce in ratione duplicata diſtantiæ *SP*. *Q.E.I.*

LIBER PRIMUS.

Idem aliter.

Cum vis ad centrum Ellipſeos tendens qua corpus *P* in Ellipſi illa revolvi poteſt, ſit (per Corol. 1 Prop. X) ut *CP* diſtantia corporis ab Ellipſeos centro *C*; ducatur *CE* parallela Ellipſeos tangenti *PR*: & vis qua corpus idem *P*, circum aliud quodvis Ellipſeos punctum *S* revolvi poteſt, ſi *CE* & *PS* concurrant in *E*, erit ut $\frac{PE\ cub.}{SP\ q.}$ (per Corol. 3. Prop. VII.) hoc eſt, ſi punctum *S* ſit umbilicus Ellipſeos, adeoque *PE* detur, ut $SP\ q.$ reciproce. *Q.E.I.*

Eadem brevitate qua traduximus Problema quintum ad Parabolam, & Hyperbolam, liceret idem hic facere: verum ob dignitatem Problematis & uſum ejus in ſequentibus, non pigebit caſus ceteros demonſtratione confirmare.

PROPOSITIO XII. PROBLEMA VII.

Moveatur corpus in Hyperbola: requiritur Lex vis centripetæ tendentis ad umbilicum figuræ.

Sunto *CA*, *CB* ſemi-axes Hyperbolæ; *PG*, *KD* diametri conjugatæ; *PF*, *Qf* perpendicula ad diametros; & *Qv* ordinatim applicata ad diametrum *GP*. Agatur *SP* fecans cum diametrum *DK* in *E*, tum ordinatim applicatam *Qv* in *x*, & compleatur parallelogrammum *QRPx*. Patet *EP* æqualem eſſe ſemiaxi tranſverſo *AC*, eo quod, acta ab altero Hyperbolæ umbilico *H* linea *HI* ipſi *EC* parallela, ob æquales *CS*, *CH*, æquentur *ES*, *EI*; adeo ut *EP* ſemidifferentia ſit ipſarum *PS*, *PI*, id eſt (ob parallelas *IH*, *PR* & angulos æquales *IPR*, *HPZ*) ipſarum *PS*, *PH*, quarum differentia axem totum 2*AC* adæquat. Ad *SP* demittatur perpendicularis *QT*. Et Hyperbolæ latere recto principali (ſeu $\frac{2BC\ quad.}{AC}$) dicto *L*, erit $L \times QR$ ad $L \times Pv$ ut *QR* ad *Pv*, id eſt, ut *PE* ſeu *AC* ad *PC*; Et $L \times Pv$ ad Gv ut *L* ad *Gv*;

G

- Was able to prove 38 of 52 theorems in *Principia*
- *“We have invented a computer program capable of thinking non-numerically, and thereby solved the venerable mind-body problem.”*
- Implemented on a RAND computer 1956

Russell and Whitehead. Principia Mathematica. 1910.

First? AI Program - Logic Theorist 1955 (Newell, Simon, Shaw)

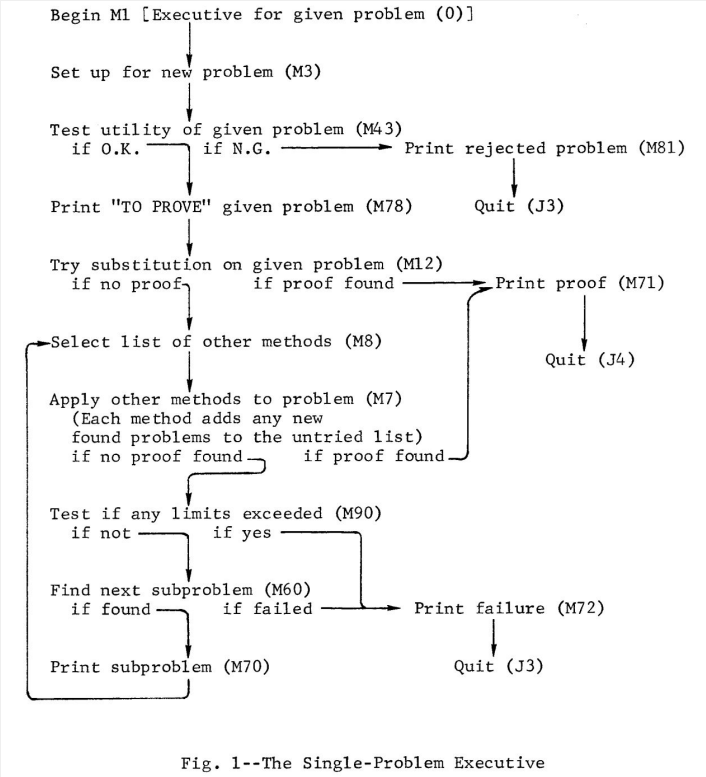
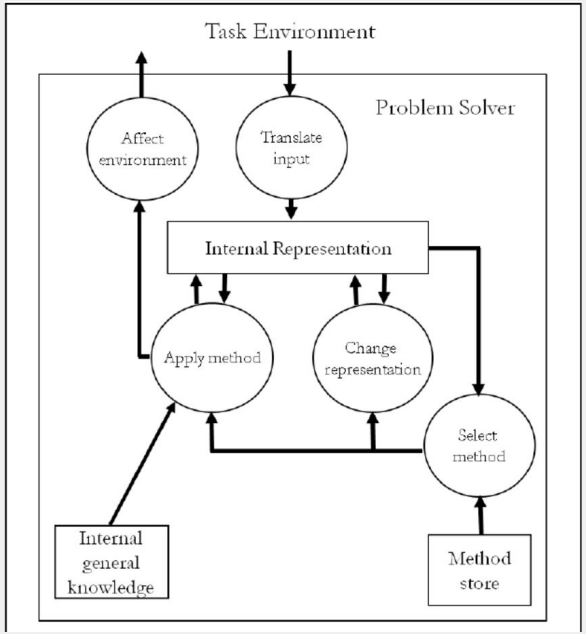


Fig. 1--The Single-Problem Executive

- Handwrote the program into index cards and then passed each card to a person who would then become a “component” (subroutine or component of its memory)
- Proving a theorem is akin to selective search on tree structure

Logic Theorist

More AI Programs 1950s



- Logic Theorist → General Problem Solver (to be more general purpose and closer to 'human thinking')
- IBM implemented the 'geometry version' of logic theorist 1959 (could "see" through punched coordinates)

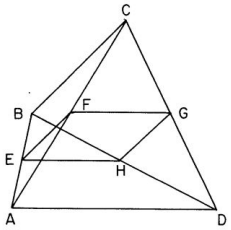
```
   PREMISES
**************
QUAD-LATERAL ABCD
POINT E MIDPOINT SEGMENT AB
POINT F MIDPOINT SEGMENT AC
POINT G MIDPOINT SEGMENT CD
POINT H MIDPOINT SEGMENT BD

   TO PROVE
**************
PARALELOGRAM EFGH

   SYNTACTIC SYMMETRIES
**************************
BA, AB, DC, CD, EE, HF, GG, FH,
CA, DB, AC, BD, GE, FF, EG, HH,
DA, CB, BC, AD, GE, HF, EG, FH,

   PROOF
***********
SEGMENT DG EQUALS SEGMENT GC
      DEFINITION OF MIDPOINT
SEGMENT CF EQUALS SEGMENT FA
      DEFINITION OF MIDPOINT
TRIANGLE DCA
      ASSUMPTION BASED ON DIAGRAM
PRECEDES DGC
      DEFINITION OF MIDPOINT
PRECEDES CFA
      DEFINITION OF MIDPOINT
SEGMENT GF PARALLEL SEGMENT AD
      SEGMENT JOINING MIDPOINTS OF SIDES OF TRIANGLE IS PARALLEL TO BASE
SEGMENT HE PARALLEL SEGMENT AD
      SYNTACTIC CONJUGATE
SEGMENT GF PARALLEL SEGMENT EH
      SEGMENTS PARALLEL TO THE SAME SEGMENT ARE PARALLEL
SEGMENT HG PARALLEL SEGMENT FE
      SYNTACTIC CONJUGATE
QUAD-LATERAL HGFE
      ASSUMPTION BASED ON DIAGRAM
PARALELOGRAM EFGH
      QUADRILATERAL WITH OPPOSITE SIDES PARALLEL IS A PARALLELOGRAM


   TOTAL ELAPSED TIME    1.03 MINUTES
```



General Problem Solver

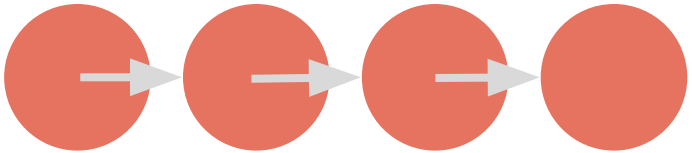
Geometry Theorem Solver

A new programming language for AI - IPL (Simon, Newell, Shaw)

- Before: FORTRAN - need to declare all arrays beforehand
- How to make lists more memory efficient and user-friendly?
- IPL - Information Processing Language mimics how the human mind works
  - thoughts are constantly created, changing, and destroying
  - links that connect symbols to other symbols in the mind can learned
- Introduced memory pointers!

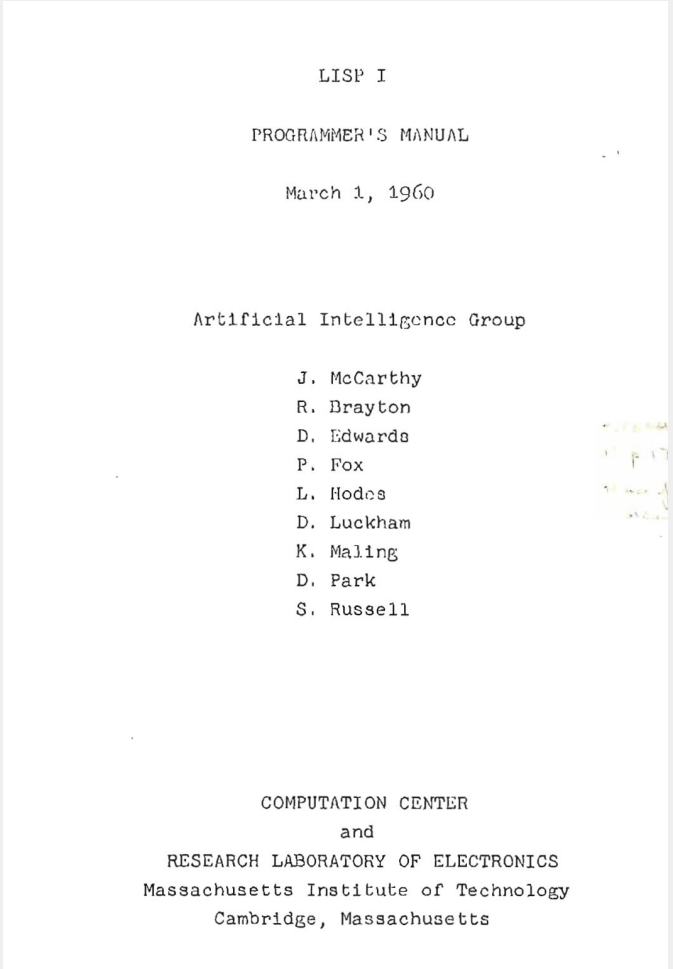
Herbert Simon describes the list-processing technique introduced in IPL

“The basic idea is that, whenever a piece of information is stored in memory, additional information should be stored with it telling where to find the next piece of information. In this way the entire memory could be organized like a long string of beads, but with the individual beads of the string stored in arbitrary locations…. “Nextness” was not determined by physical propinquity but by an address or **pointer** stored with each item.” - Herbert Simon



IPL inspires LISP I 1960 (John McCarthy)

- LISP made it much easier to insert, delete memory at any point in the list
- “programming language of AI” for the next 30 years
- LISP also introduced the idea of Lambda-functions where each function could be used as an object
- But required a compiling process in lower level languages that took long!



LISP I

PROGRAMMER'S MANUAL

March 1, 1960

Artificial Intelligence Group

J. McCarthy
R. Brayton
D. Edwards
P. Fox
L. Hodes
D. Luckham
K. Maling
D. Park
S. Russell

COMPUTATION CENTER
and
RESEARCH LABORATORY OF ELECTRONICS
Massachusetts Institute of Technology
Cambridge, Massachusetts

Advice Taker need creates LISP

“Apart from consulting on the geometry program, my own research in artificial intelligence was proceeding along the lines that led to the **Advice Taker** proposal in 1958 (McCarthy 1959). This involved representing information about the world by sentences in a suitable formal language and a reasoning program that would decide what to do by making logical inferences. Representing sentences by list structure seemed appropriate - it still is - and a list processing language also seemed appropriate for programming the operations involved in deduction - and still is.”

John McCarthy “LISP prehistory - Summer 1956 through Summer 1958.” 1979.

6.1 General Description

(1) Representation of S-expressions by List Structure

A list structure is a collection of computer words arranged as in Fig. 1a or 1b. Each word of the list structure is represented by one of the subdivided rectangles in the figure. The left box of a rectangle represents the address field of the word and the right box represents the decrement field. An arrow from a box to another rectangle means that the field corresponding to the box contains the location of the word corresponding to the other rectangle.

It is permitted for a substructure to occur in more than one place in a list structure, as in Fig. 1b, but it is not permitted for a structure to have cycles, as in Fig. 1c.

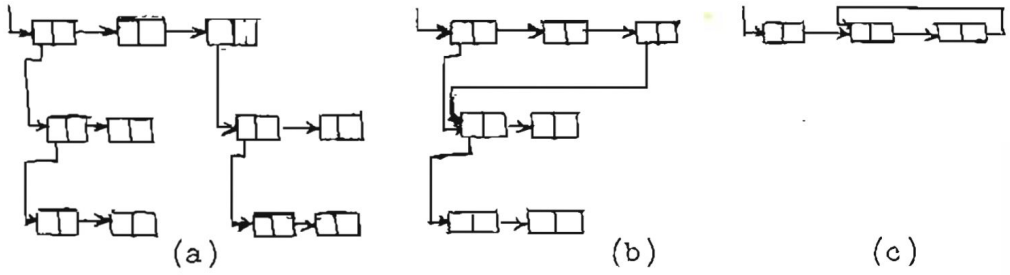


Fig. 1

The advantages of list structures for the storage of symbolic expressions are:

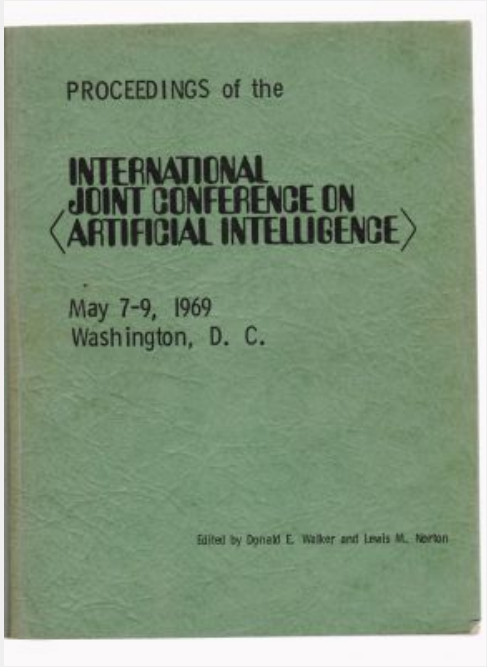
1. The size and even the number of expressions with which the program will have to deal cannot be predicted in advance. Therefore, it is difficult to arrange blocks of storage of fixed length to contain them.

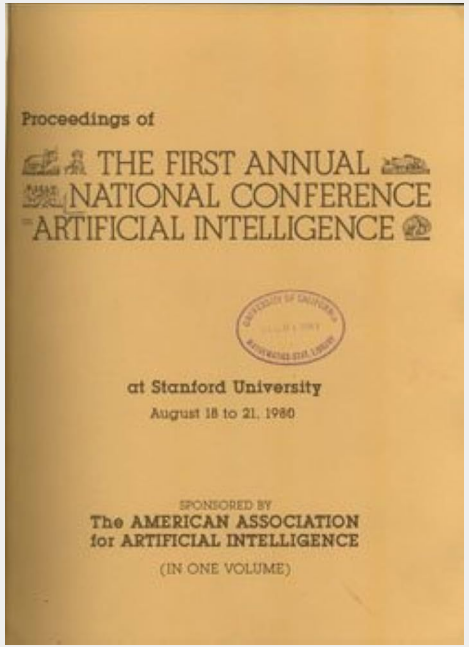
2. Registers can be put back on the free-storage list when they are no longer needed. Even one register returned to the list is of value, but if expressions are stored linearly, it is difficult to make use of blocks of registers of odd sizes that may become available.

3. An expression that occurs as a subexpression of several expressions need be represented in storage only once.

(more) AI Conferences & Expos

International Joint Conference on Artificial Intelligence (IJCAI) 1969







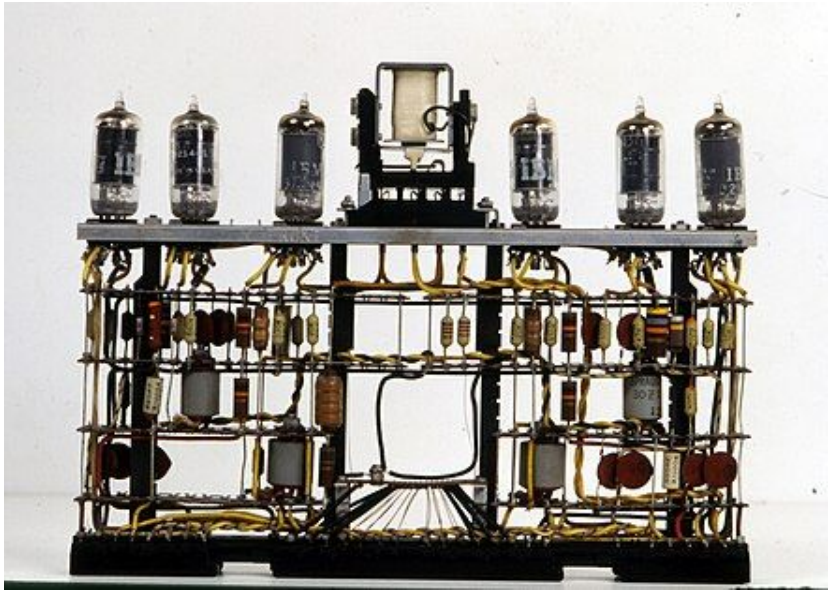
Consumer Electronics Show 1967

AAAI Conference (1980)

European Conference on AI (ECAI) [previously AISB] 1974

Meanwhile, IBM

Introduced in 1954, IBM 704 was the first mass produced computer with floating-point arithmetic capacity



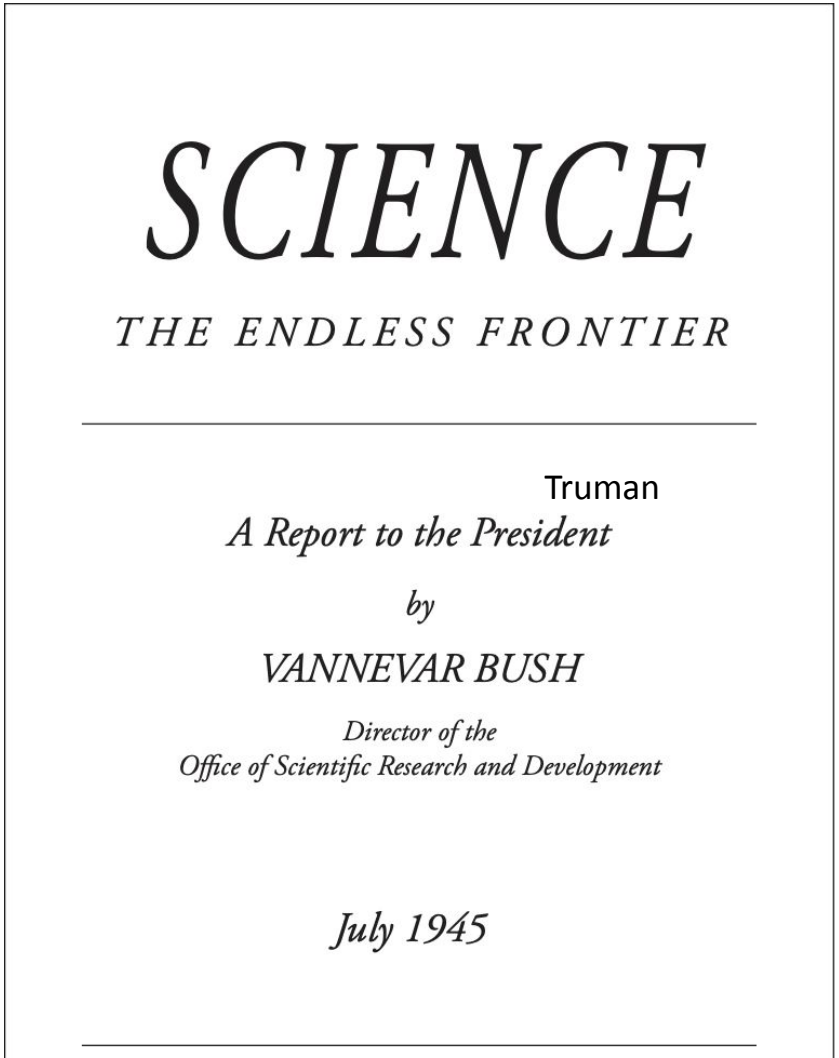
Vacuum Tubes



Arthur Samuel in 1952 wrote programs for the IBM 704 playing checkers at “strong amateur” level



Cold War!



U.S. funds

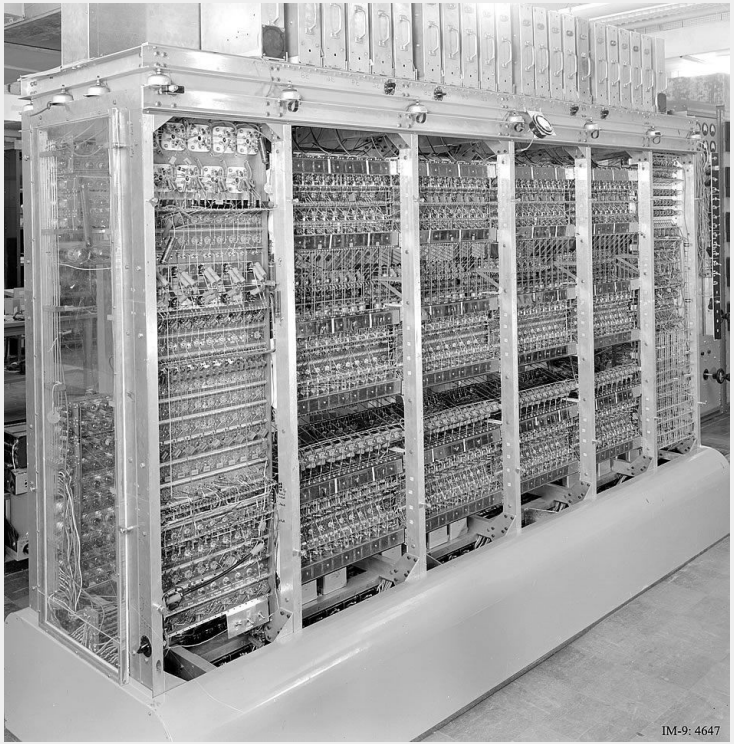
"War is increasingly total war, in which the armed services must be supplemented by active participation of every element of the civilian population.

There must be more - and more adequate - military research during peacetime….The obligation of Government to support research on military problems is inescapable.

Modern war requires the use of the most advanced scientific techniques. Many of the leaders in the development of radar are scientists who before the war had been exploring the nucleus of the atom. While there must be increased emphasis on science in the future training of officers for both the Army and Navy, such men cannot be expected to be specialists in scientific research. Therefore a professional partnership between the officers in the Services and civilian scientists is needed."

ENIAC → MANIAC 1946

- Mathematical Numerical Integrator Computer
- Von Neumann funded by Atomic Energy Commission for Hydrogen bomb research
- Smarter and smaller than ENIAC
- “Computer and the Brain” 1957:
  - “artificial automaton that has been constructed for human use.”



1 March, 1954

- Castle Bravo
- First in the series of hydrogen bomb tests
- Bikini Atoll, Marshall Islands
- “With a predicted yield of 6 megatons, Castle Bravo would deliver twice as much power as all the bombs dropped on Germany and Japan during World War II together, including both atomic bombs.”
- Radiation fallout in nearby countries



October 4, 1957

"All the News That's Fit to Print"

The New York Times.

LATE CITY EDITION

VOL. CVII.. No. 36,454. NEW YORK, SATURDAY, OCTOBER 5, 1957. FIVE CENTS

SOVIET FIRES EARTH SATELLITE INTO SPACE; IT IS CIRCLING THE GLOBE AT 18,000 M. P. H.; SPHERE TRACKED IN 4 CROSSINGS OVER U. S.

HOFFA IS ELECTED TEAMSTERS' HEAD; WARNS OF BATTLE

Defeats Two Foes 3 to 1 —Says Union Will Fight 'With Every Ounce'



IN TOKEN OF VICTORY: Dave Beck, retiring head of the Teamsters Union, raises hand of James R. Hoffa upon his election as union's president. At right is Mrs. Hoffa.

COURSE RECORDED

Navy Picks Up Radio Signals—4 Report Sighting Device



Device Is 8 Times Heavier Than One Planned by U.S.

560 MILES HIGE

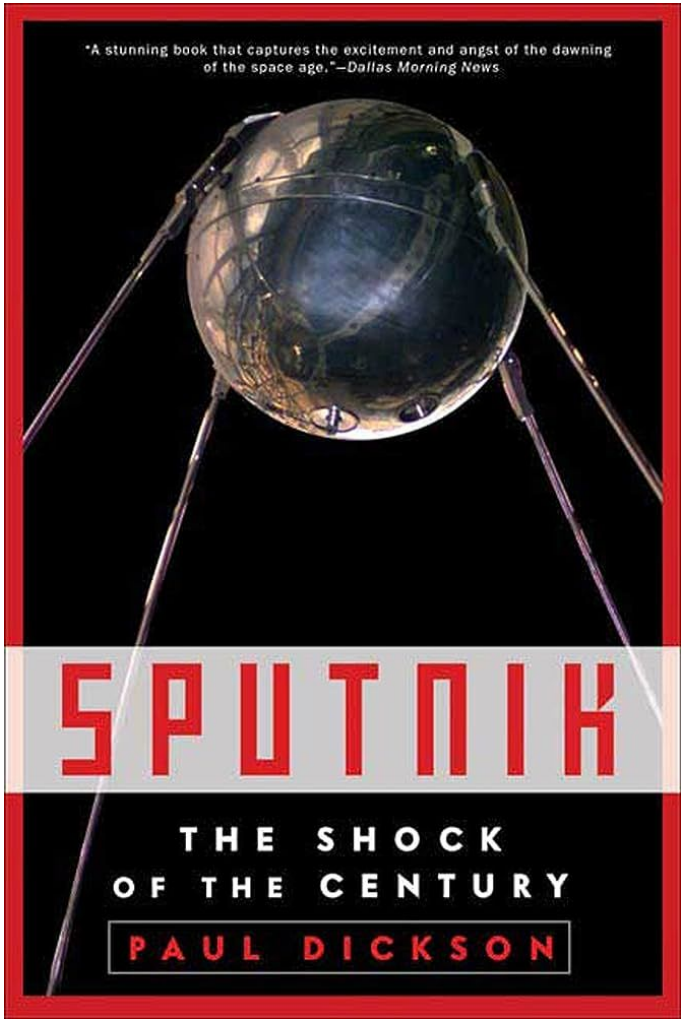
Visible With Simple Binoculars, Moscow Statement Says

FAUBUS COMPARES HIS STAND TO LEE'S

Flu Widens in City; 10% Rate Predicted; 200,000 Pupils Out

ARGENTINA TAKES EMERGENCY STEPS

SATELLITE SIGNAL



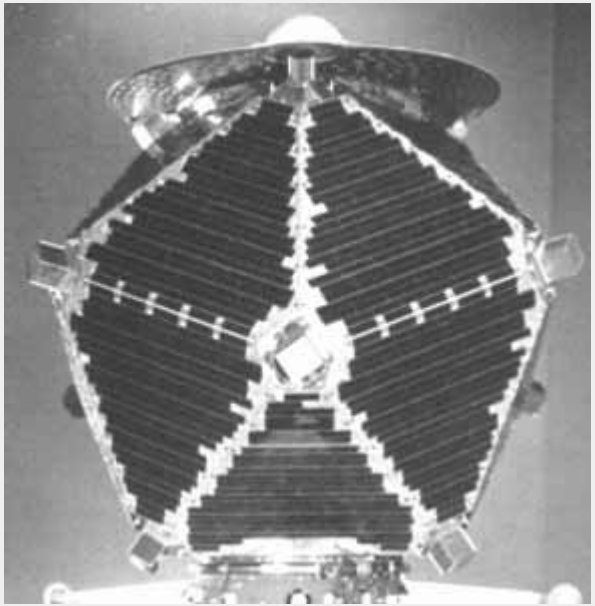
January 10th 1958
ARPA - rise of interactive computing for defense

- Eisenhower administration creates Advanced Research Projects Agency
- Until NASA was founded, originally an interim space agency (under Office of Secretary of Defense)
- Research = Security
- Was able to sponsor high risk projects, without congressional oversight, not subject to peer-review



ARPA & MAD

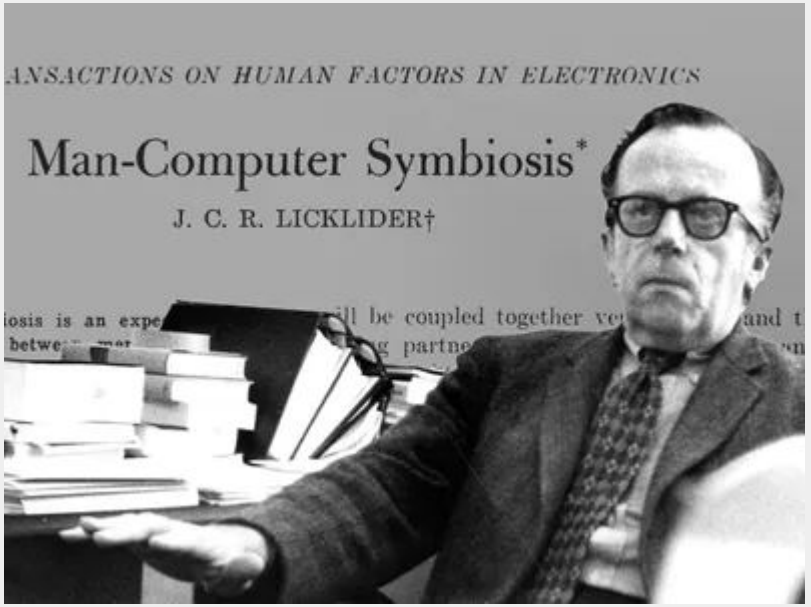
- Vela: Seismic & Atmospheric Sensing (Partial Nuclear Test Ban)
- Defender: Antiballistic Missile Project



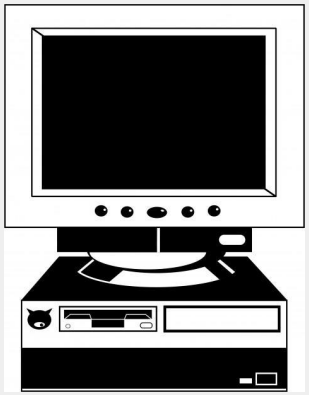
Vela Satellite

ARPA's IPTO 1961
Information Processing Techniques Office

- JFK's directive to step up "command and control" research
- Headed by "Lick" J.C.R. Licklider, psychologist who studied computer interaction & psychoacoustics in the 1950s
- "the electronic members of the partnership"
- **"Command and control" included research in ballistic missiles, radar tracking, high-energy lasers, etc.**



“Lick’s” Man-Computer Symbiosis 1960





speed and accuracy

flexibility and intuition

“The hope is that, in not too many years, human brains and computing machines will be coupled together very tightly, and that the resulting partnership will think as no human brain has ever thought and process data in a new way not approached by the information-handling machines we know today.”

JCR Licklider, Man-Computer Symbiosis, 1960

JCR Licklider



Time & Language: 2 components of Symbiosis

1. **Time**

   Batch-processing was too slow → need time sharing for real-time, interactive computing (especially in command-and-control issues)

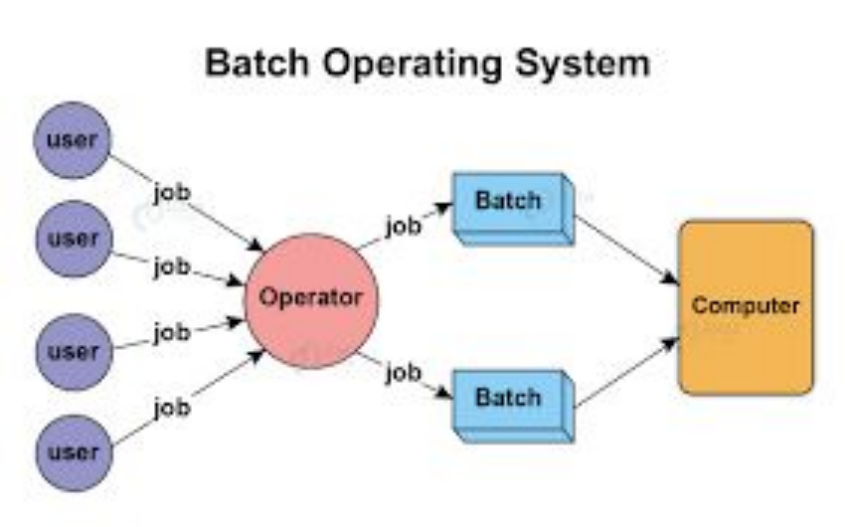
2. **Language**

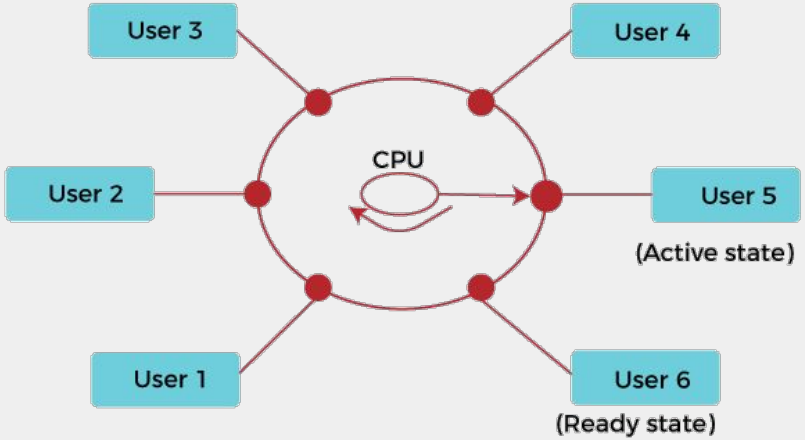
   Assembly → programming languages more similar to human beings (IPL/LISP); Computers and humans should be able to interact in ordinary speech (especially in military command-and-control issues)

It is to bring computing machines effectively into processes of thinking that must go on in **"real time,"** time that moves too fast to permit using computers in conventional ways. Imagine trying, for example, to direct a battle with the aid of a computer on such a schedule as this. You formulate your problem today. Tomorrow you spend with a programmer. Next week the computer devotes 5 minutes to assembling your program and 47 seconds to calculating the answer to your problem. You get a sheet of paper 20 feet long, full of numbers that, instead of providing a final solution, only suggest a tactic that should be explored by simulation. Obviously, the **battle would be over before the second step in its planning was begun.** To think in interaction with a computer in the same way that you think with a colleague whose competence supplements your own will require much tighter coupling between man and machine than is suggested by the example and than is possible today.

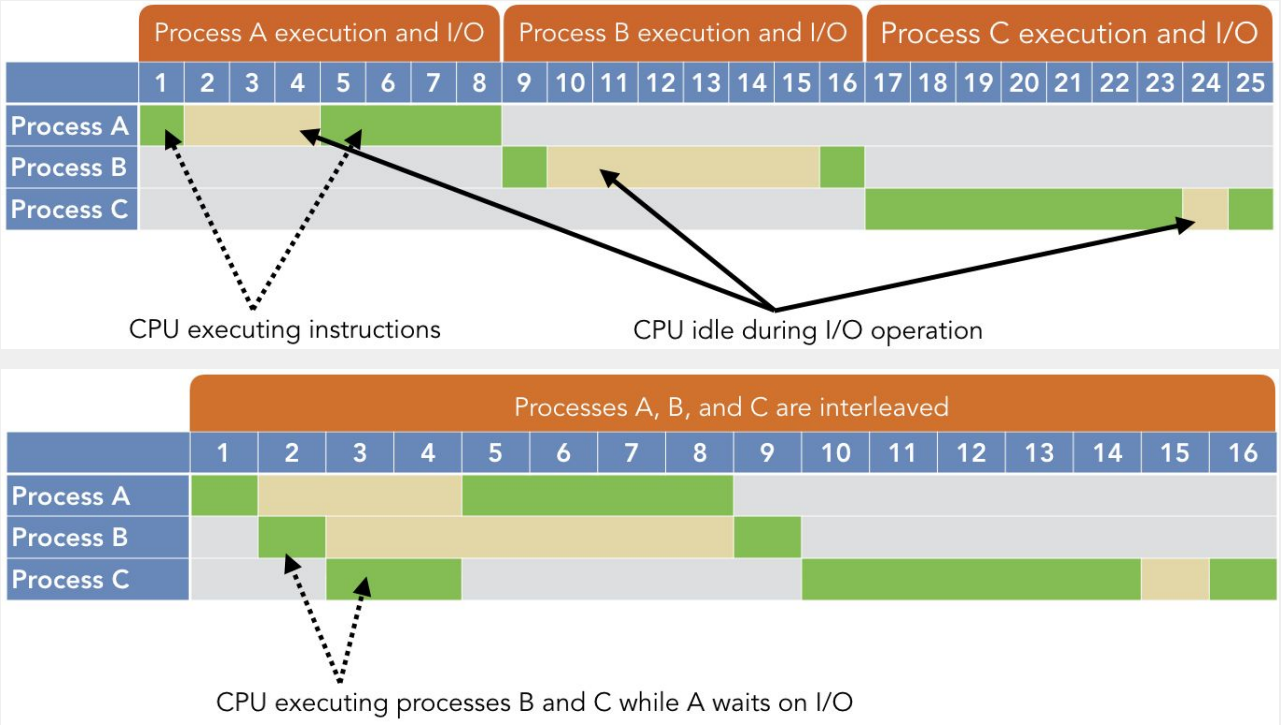
The basic **dissimilarity between human languages and computer languages may be the most serious obstacle to true symbiosis**. It is reassuring, however, to note what great strides have already been made, through interpretive programs and particularly through assembly or compiling programs such as FORTRAN, to adapt computers to human language forms. The "Information Processing Language" of Shaw, Newell, Simon, and Ellis [24] represents another line of rapprochement.

Batch Processing vs. Time-Sharing



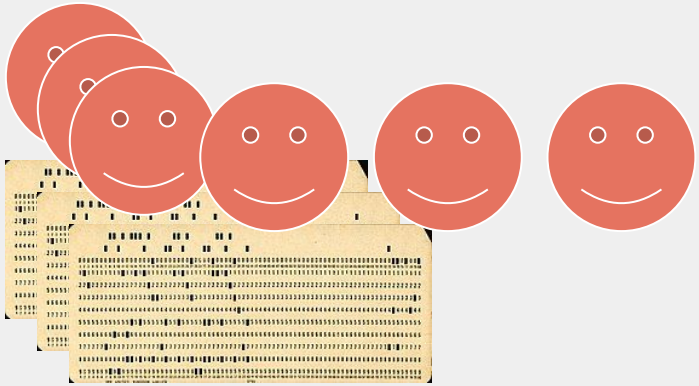


(Multiprogramming)

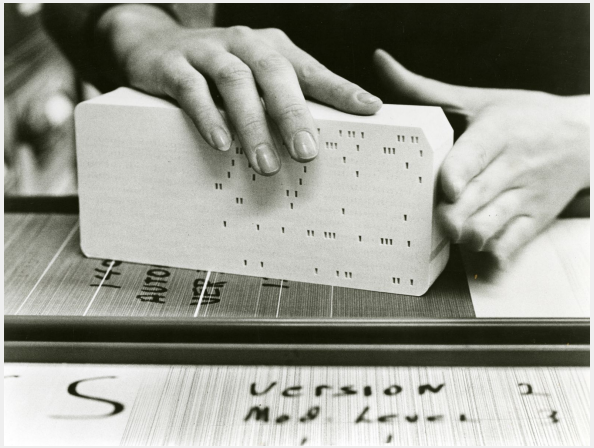


https://w3.cs.jmu.edu/kirkpams/OpenCSF/Books/csf/html/Multiprogramming.html

From Batch Processing …



Jobs



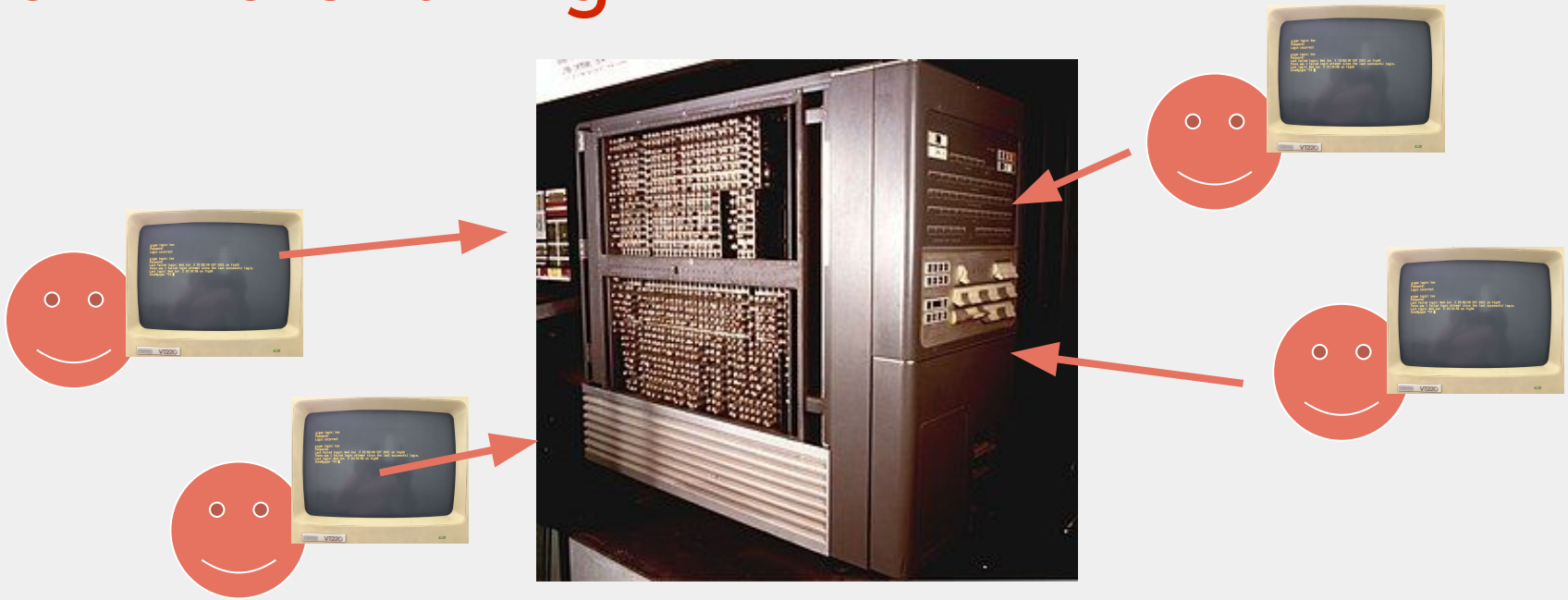
Batch



Processed

… to Time-Sharing

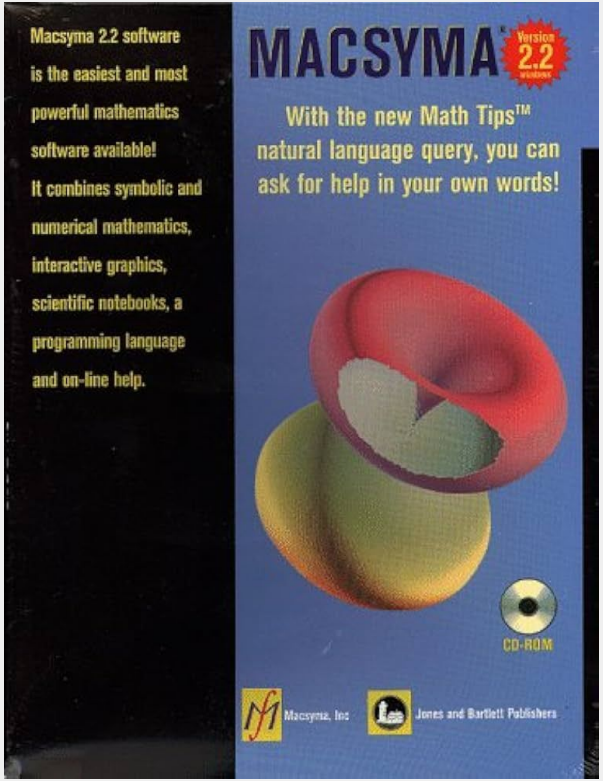
Illusion of real-time computing



Interactive Computing out of IPTO

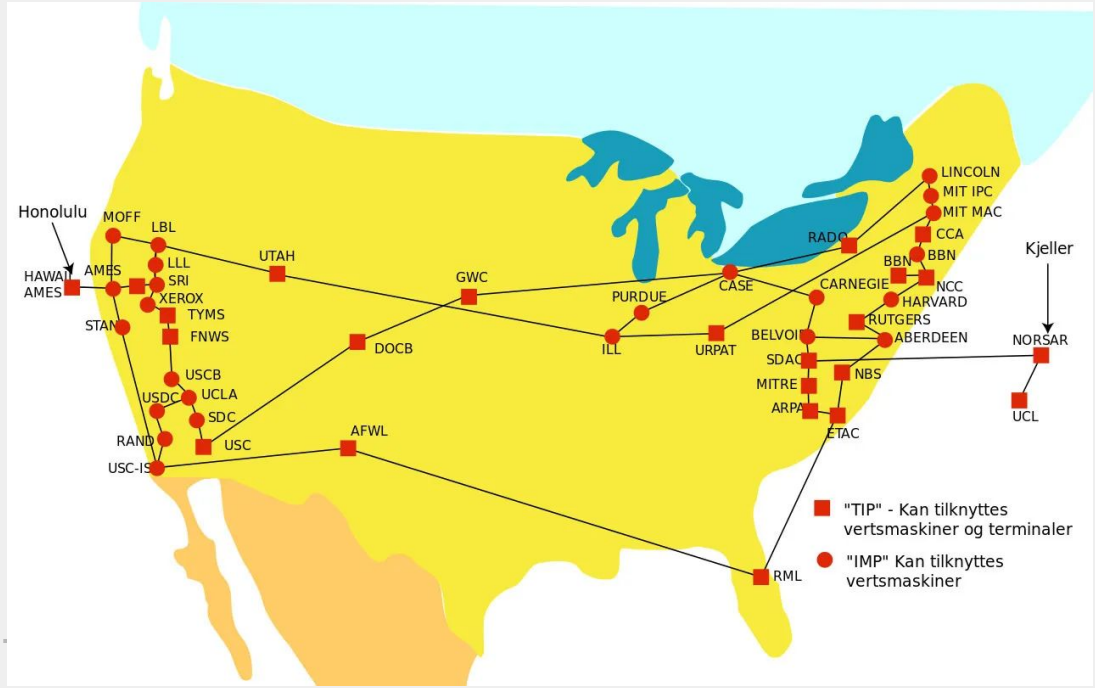
- MAC (Man and Computer; Multi-Access Computing): ARPA spent $25m 1964-1970
- Multics, TSS (Time sharing interfaces)
- MACSYMA (High level algebra program)

```
$TIME SHARING SYSTEM ON THE AIR
LOGIN 10335 JG025
$OK LOG ON 16
DRUMS
$4324 DRUM WDS.
DIAL 9  THIS IS JOHN JONES, I NEED 20K IN ORDER TO LOAD MY PROG.
$MSG IN
FROM 9  WE CAN GET YOU ON IN 5 MINUTES.
FROM 9  GO AHEAD AND LOAD.
LOAD ALPHA1
$LOAD OK
(CHANNEL 9'S TELETYPE COPY - THE OPERATOR'S TELETYPE)
$TIME SHARING SYSTEM ON THE AIR
LOGIN 8762 JG025 MANAGR  (LOAD REQUEST FOR MANAGER PROGRAM)
$LOAD OK
GO       (OPERATOR TYPES 'GO' TO START PROGRAM)
ENTER SCOPE NO.  6  (PROG. QUERIES OPERATOR FOR SCOPE NO.)
AM,20    (REQUESTS MANAGER PROG. TO UPDATE SCOPE EVERY 20 SEC.)
FROM 16  THIS IS JOHN JONES, I NEED 20K IN ORDER TO LOAD MY PROG.
26   (REQUESTS MANAGER PROG. TO GIVE A DETAILED DISPLAY OF CH. 26)
!DIAL 26  CAN YOU FINISH IN 5 MIN. - WE HAVE A HIGH PRIORITY REQUEST.
$MSG IN
FROM 26  OK - I'LL FINISH UP IN 5 MINUTES.
DIAL 16  WE CAN GET YOU ON IN 5 MINUTES.
$MSG IN
FROM 26  I'M THROUGH
OT 26   (OPERATOR QUITS CHANNEL 26)
DIAL 16  GO AHEAD AND LOAD.
$MSG IN
```



“Intergalactic Computer Network”

- ARPANET → MILNET → Internet
- “Libraries of the Future”



Vietnam War & Project Agile



- Seismic Sensors
- Magnetic Sensors
- Infrared Sensors
- Strain Sensors

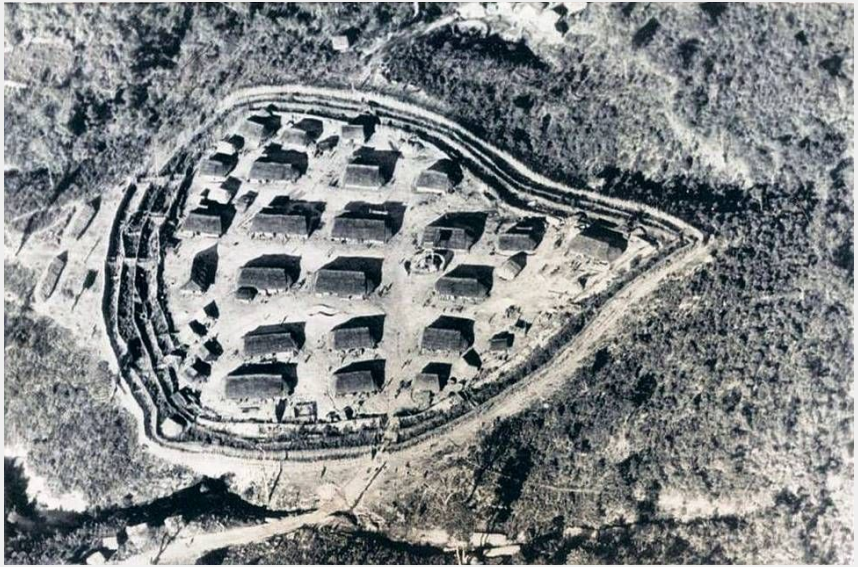
Sensors dropped on Ho Chi Minh Trail



First Drones. 1968

Behavioral Experiments

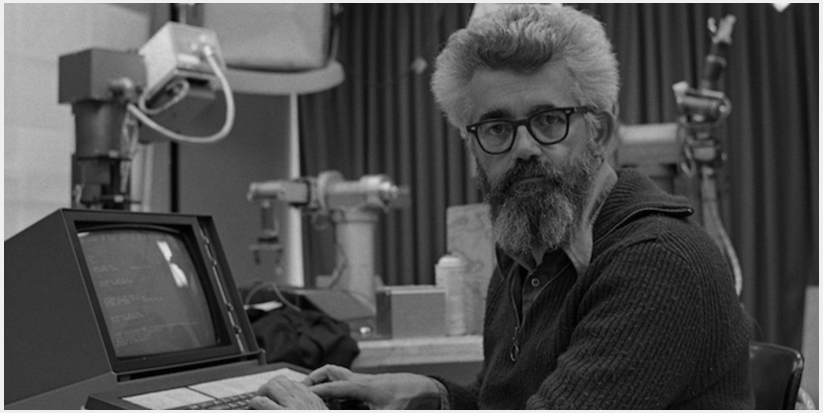
- “Thailand was the laboratory for the soft side and Vietnam was the laboratory for the hard side, or things that go boom”
- “Anthropometric Survey of the Royal Thai Armed Forces” - 2,950 Thai soldiers, sailors, and pilots
- Strategic Hamlet Program to isolate insurgency in Vietnam 1964
- All data taken back to Washington and computed



Strategic Hamlet

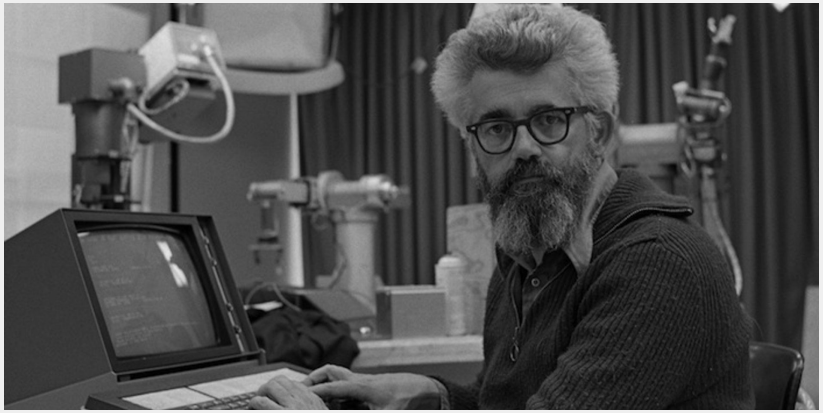
ARPA funds U.S. AI institutions

- MIT: Marvin Minsky and students
- Stanford: John McCarthy started the Stanford AI Lab with unconditional ARPA money 1964
- ARPA supported AI at Rand, Carnegie-Mellon, SRI, SDC (Systems Development Corp) ,BBN (Bolt, Beranak, Newman), and seven other “centers of excellence”
- ARPA became the primary patron for the first twenty years of AI research



ARPA funds U.S. AI institutions

- By the mid-1970s, IPTO's budget accounted for 40-50% of applied research in math and computers
- 80-90% of funding for major AI groups (MIT, Stanford, SRI, CMU) came from ARPA/DARPA



ANALOGY 1963 (Tom Evans)

- Rule-based program written in LISP
- No scanners → coordinate points entered manually
- **'figure A is to figure B as figure C is to which of the given answer figures?'**

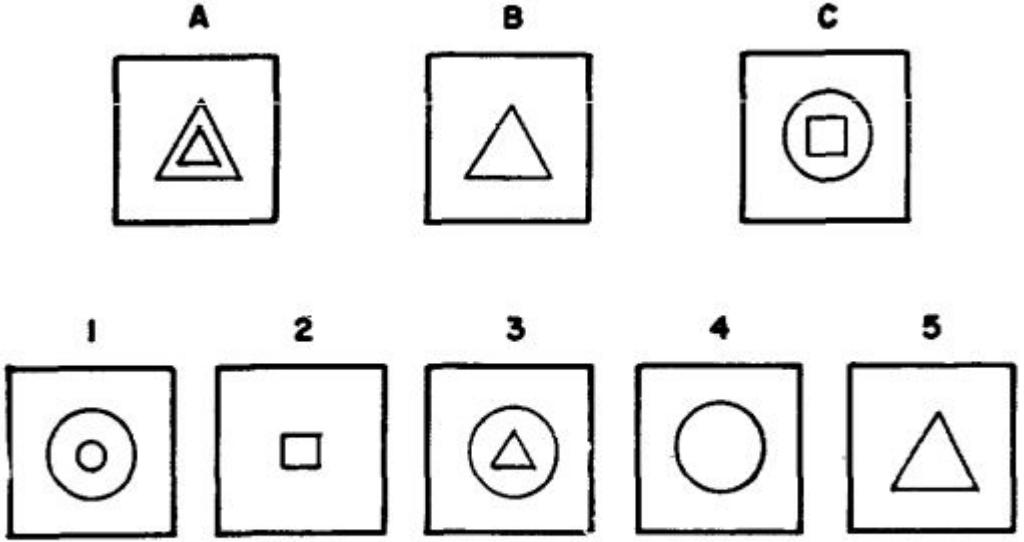


Figure 1.

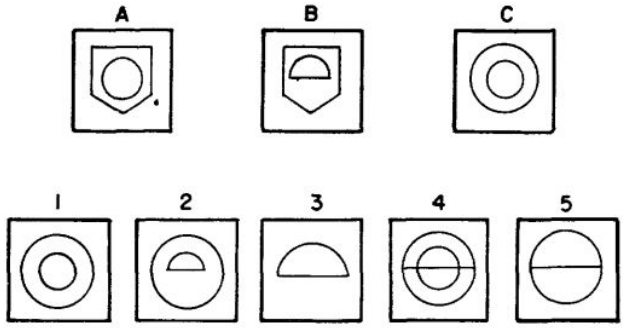


Figure 6.

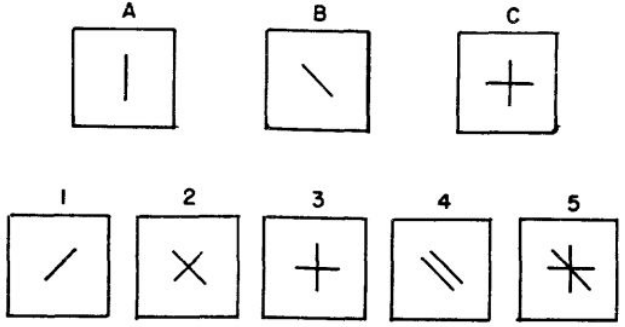


Figure 8.

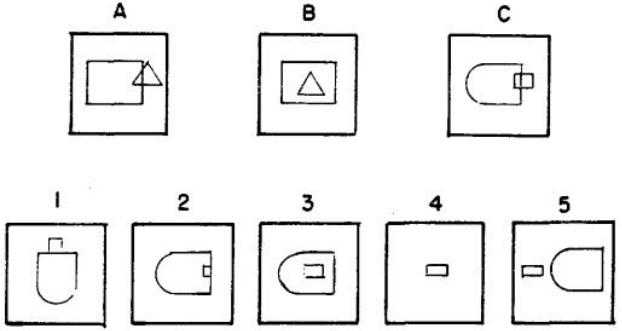


Figure 7.

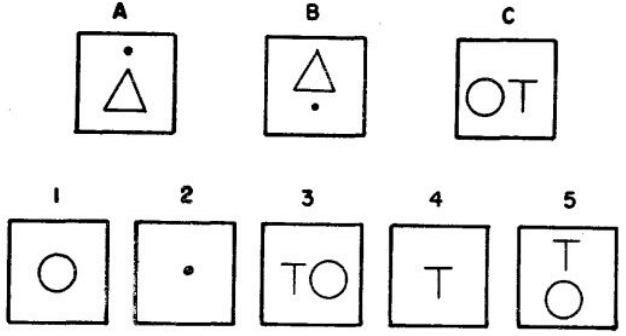


Figure 9.

```
(
(REMOVE A1 ((ABOVE A1 A3) (ABOVE
    A1 A2) (SIM OB3 A1 (((1.0 . 0.0) .
    (N.N))))))
(MATCH A2 (((INSIDE A2 A3) (ABOVE
    A1 A2)) . ((LEFT A2 A3) (SIMTRAN
    (((1.0 . 0.0) . (N.N)) ((1.0 . 3.14) .
    (N.N)))))))
(MATCH A3 (((INSIDE A2 A3) (ABOVE
    A1 A3)) . ((LEFT A2 A3) (SIMTRAN
    (((1.0 . 0.0) . (N.N)))))))
)
```

STUDENT 1964 (Daniel Bobrow)

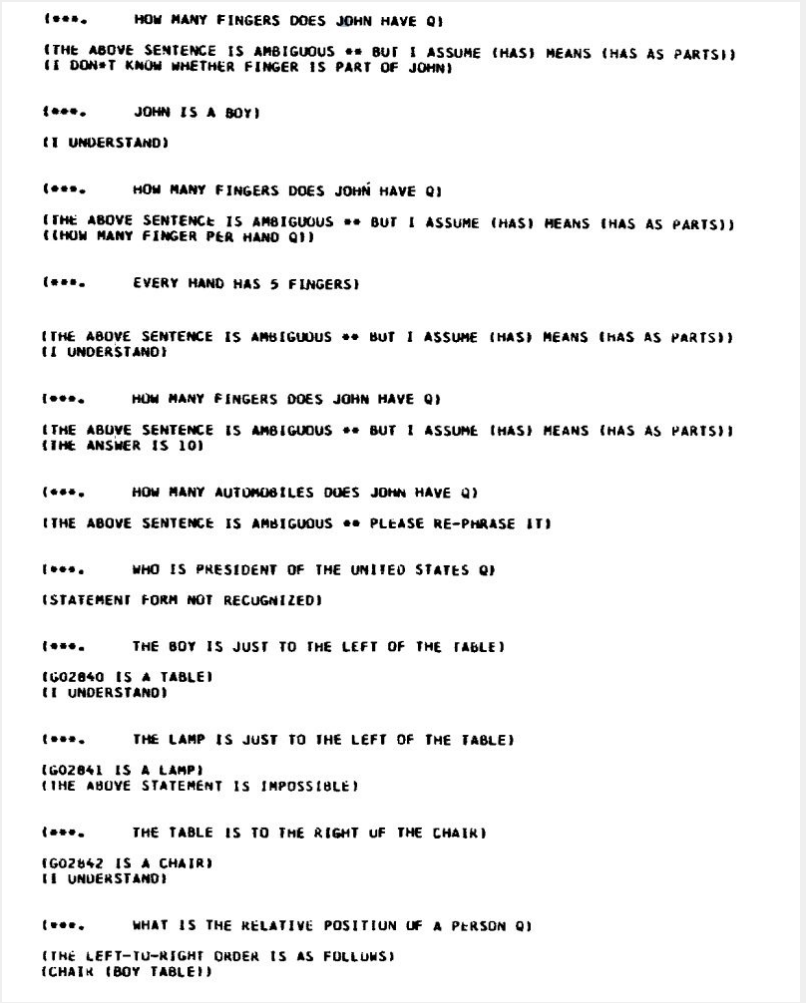
- Rule-based program written in LISP
- Solve math word problems at high school level
- English sentences are converted into kernel sentences (single pieces of info) → mathematical expressions. 52 facts supported.

    - **“Bill’s father’s uncle is twice as old as Bill’s father. Two years from now Bill’s father will be 3 times as old as Bill. The sum of their ages is 92. Find Bill’s age.”**

    - **“If the number of customers Tom gets is twice the square of 20% of the number of advertisements he runs, and the number of advertisements is 45, then what is the number of customers Tom gets?”**

SIR 1964 (Bertram Raphael)

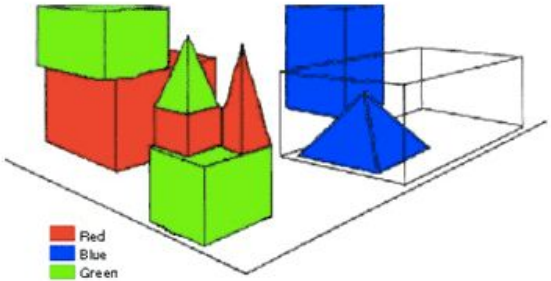
- Rule-based program written in LISP
- Semantic Information Retrieval
- Program interpreted or “understood” simple English relations like ownership, spatial relations, part-whole relations.
- **It used “template matching” to “learn” facts and retrieve them.**
  - **the sum of _____ and _____**



Microworlds 1972

- Real world has too many variables → reduce them to a artificial simulation!
- SHRDLU was a rule-based program made to understand natural language
- The program could have natural language conversations about the blocks in block world and manipulate them
- Researchers hoped that by expanding this microworld, more comprehensive AI could emerge





Person: Pick up a big red block.

Computer: OK.

Person: Grasp the pyramid.

Computer: I don't understand which pyramid you mean.

Person: PICK UP A BIG RED BLOCK.
Computer: OK. (does it)
Person: GRASP THE PYRAMID.
Computer: I DON'T UNDERSTAND WHICH PYRAMID YOU MEAN.
Person: FIND A BLOCK WHICH IS TALLER THAN THE ONE YOU ARE HOLDING AND PUT IT INTO THE BOX.
Computer: BY "IT", I ASSUME YOU MEAN THE BLOCK WHICH IS TALLER THAN THE ONE I AM HOLDING.
Computer: OK. (does it)
Person: WHAT DOES THE BOX CONTAIN?
Computer: THE BLUE PYRAMID AND THE BLUE BLOCK.
Person: WHAT IS THE PYRAMID SUPPORTED BY?
Computer: THE BOX.
Person: HOW MANY BLOCKS ARE NOT IN THE BOX?
Computer: FOUR OF THEM.
Person: IS AT LEAST ONE OF THEM NARROWER THAN THE ONE WHICH I TOLD YOU TO PICK UP?
Computer: YES, THE RED CUBE.
Person: IS IT SUPPORTED?
Computer: YES, BY THE TABLE.
Person: CAN THE TABLE PICK UP BLOCKS?
Computer: NO.
Person: CAN A PYRAMID BE SUPPORTED BY A BLOCK?
Computer: YES.
Person: CAN A PYRAMID SUPPORT A PYRAMID?
Computer: I DON'T KNOW.
Person: STACK UP TWO PYRAMIDS.
Computer: I CAN'T.
Person: WILL YOU PLEASE STACK UP BOTH OF THE RED BLOCKS AND EITHER A GREEN CUBE OR A PYRAMID?
Computer: OK. (does it)
Person: WHICH CUBE IS SITTING ON THE TABLE?
Computer: THE LARGE GREEN ONE WHICH SUPPORTS THE RED PYRAMID.
Person: IS THERE A LARGE BLOCK BEHIND A PYRAMID?
Computer: YES, THREE OF THEM: A LARGE RED ONE, A LARGE GREEN CUBE, AND THE BLUE ONE.
Person: PUT A SMALL ONE ONTO THE GREEN CUBE WHICH SUPPORTS A PYRAMID.
Computer: OK. (does it)
Person: PUT THE LITTLEST PYRAMID ON TOP OF IT.
Computer: OK. (does it)
Person: DOES THE SHORTEST THING THE TALLEST PYRAMID'S SUPPORT SUPPORTS SUPPORT ANYTHING GREEN?
Computer: YES, THE GREEN PYRAMID.
Person: WHAT COLOR IS IT?
Computer: BY "IT", I ASSUME YOU MEAN THE SHORTEST THING THE TALLEST PYRAMID'S SUPPORT SUPPORTS.
Computer: RED
Person: HOW MANY THINGS ARE ON TOP OF GREEN CUBES?
Computer: I'M NOT SURE WHAT YOU MEAN BY "ON TOP OF "IN THE PHRASE "ON TOP OF GREEN CUBES " .
DO YOU MEAN:

WHY DID YOU DROP IT?

BY "IT", I ASSUME YOU MEAN THE SUPERSTICK.
I CAN'T EXPLAIN A NON-EXISTENT EVENT.



PLANNER language

- LISP based
- SHRDLU written in PLANNER
- Automate tedious programming tasks like looping and backtracking (one statement was enough to get the program to loop through all until it found a block)

```
(GOAL (IS ?X1 BLOCK))
(GOAL (COLOR-OF ?X1 RED))
(GOAL (CUBE ?X1))
(GOAL (IS ?X2 PYRAMID))
```

STRIPS 1969 (SRI)

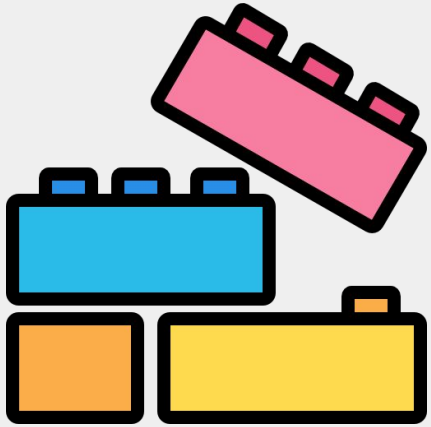
- Shakey the robot
- TV camera, range finder, radio-link antenna
- real life “micro world” of 7 rooms connected by 8 doors
- some rooms were boxes that Shakey could stack or push around
- The control procedure that guided Shakey was called STRIPS (Stanford Research Institute Problem Solver)



Rule-based Shortcomings!

- Rule-based “toy problem” microworlds did not scale
- Language translation hit a ceiling
- Dictionary based machine translation was insufficient for fluent translation. Programs needed encyclopedia knowledge to understand natural language but there was neither software nor hardware to carry this capacity in the 1960s → **In 1966, all U.S. gov support for machine translation ended.**

ENG: “the spirit is willing but the flesh is weak”
RUS: “the vodka is strong but the meat is rotten”



ARPA → DARPA 1972-

- Vietnam War increasingly unpopular (1975 “collapse” of S. Vietnam)
- Congress pressures DoD to be more frugal and spend on defense-related projects
- Greater accountability
- Less freedom/autonomy :(
- But DARPA funding doubled 1977



ARPA → DARPA money

ARPA

- IPTO program managers solicited proposals in the research community by direct contact with the researchers
- Community/Small Circle
- More experimental
- Network/Trust-oriented

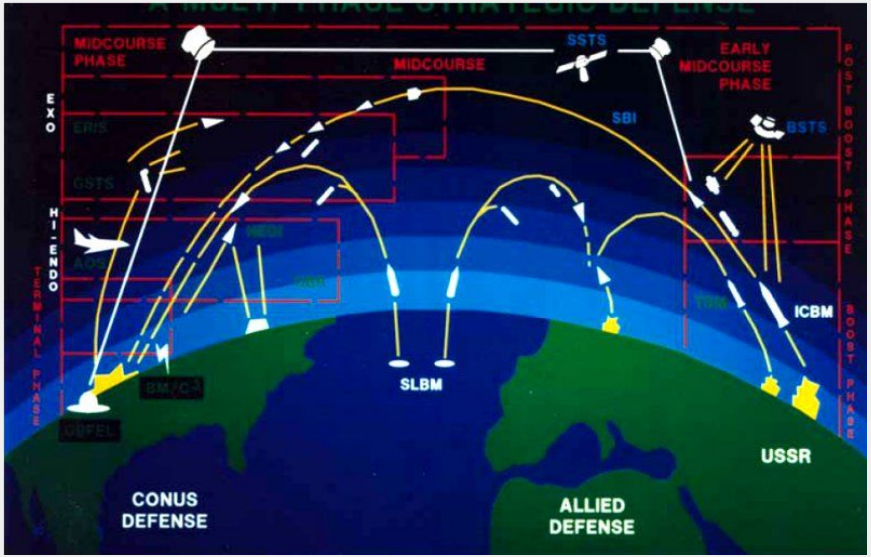
DARPA

- as the funds got larger ($100 million a year), fund granting became more official and publicized. PMs would specify project needs as much as possible.
- Slower
- More competition
- Direct relation to defense

Star Wars 1983

- Reagan announces “Star Wars” Strategic Defense Initiative for ballistic missile research and development
- 20-100 billion dollars spent



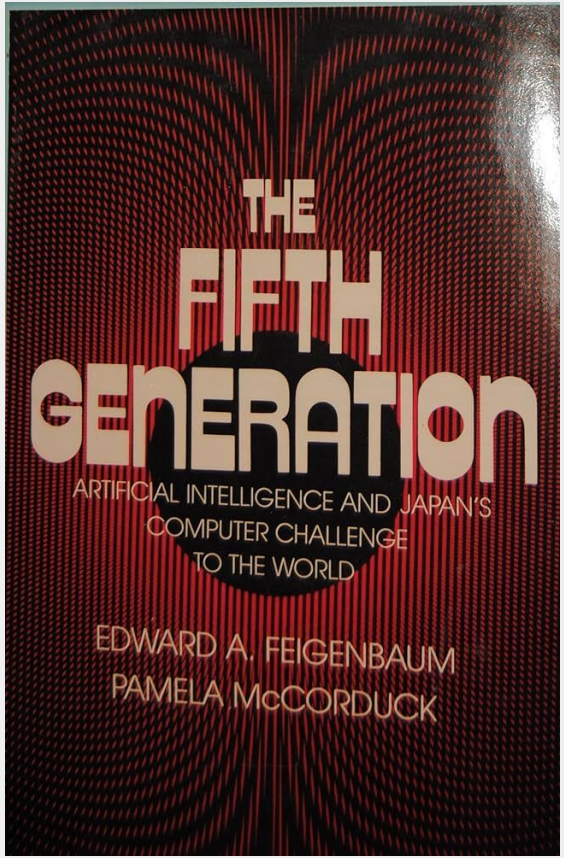


3 week later… Strategic Computing Initiative

- 1 decade long (1983-93)
- 1 billion dollars
- Inspired/threatened by Japan's (MITI's) Fifth Generation Computer Program (1982-, $800million dollar budget)
- By 1985, DARPA had spent $100 million for 92 projects underway at 60 institutions (half in industry, half in universities and gov)
- SCI planners made a detailed ten-year timeline to coordinate all of these subprojects to come together to create 3 intelligent projects

SC

STRATEGIC COMPUTING



Project 1) Autonomous Land Vehicle Project (CMU) 1984-1988

- An autonomous land vehicle (ALV) would be required to navigate unfamiliar terrain at high speeds → possibly a robot tank.
- Funded advances in machine vision

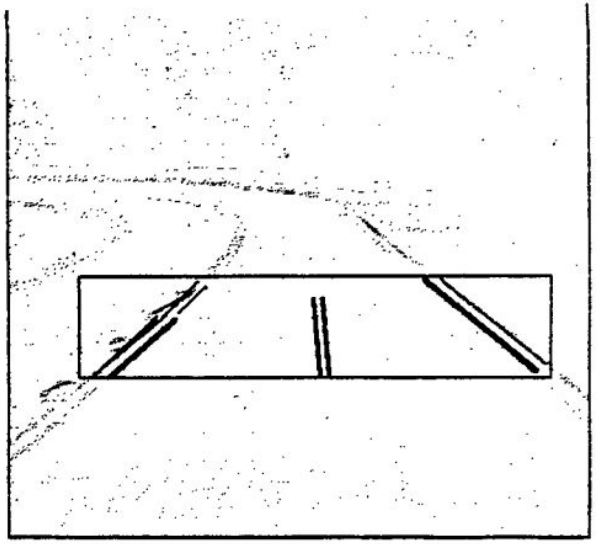


Figure 2: Road edge detection. Here a road image is shown after processing to enhance intensity changes. The vision program selects a window in which to search for road edges. Candidate left and right road edges are lines fit to the raw edge data, shown here as black lines. Heavy black lines indicate the left and right road edges selected by the program. The computed road center line is shown as a double line



Figure 8: An ERIM range image (top) and a color image (bottom) taken in registration

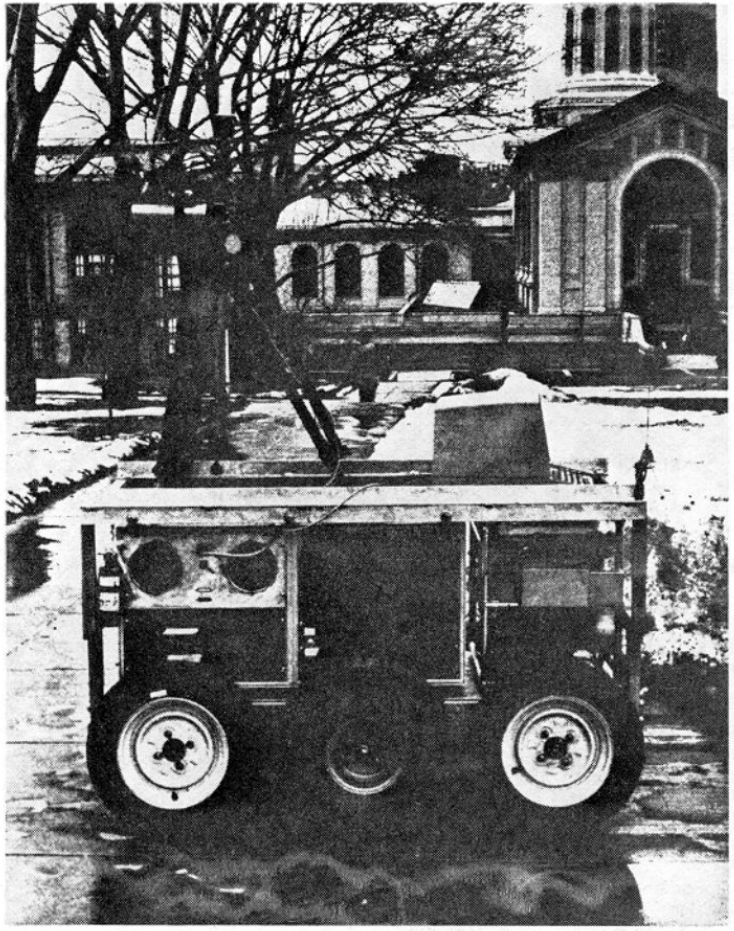
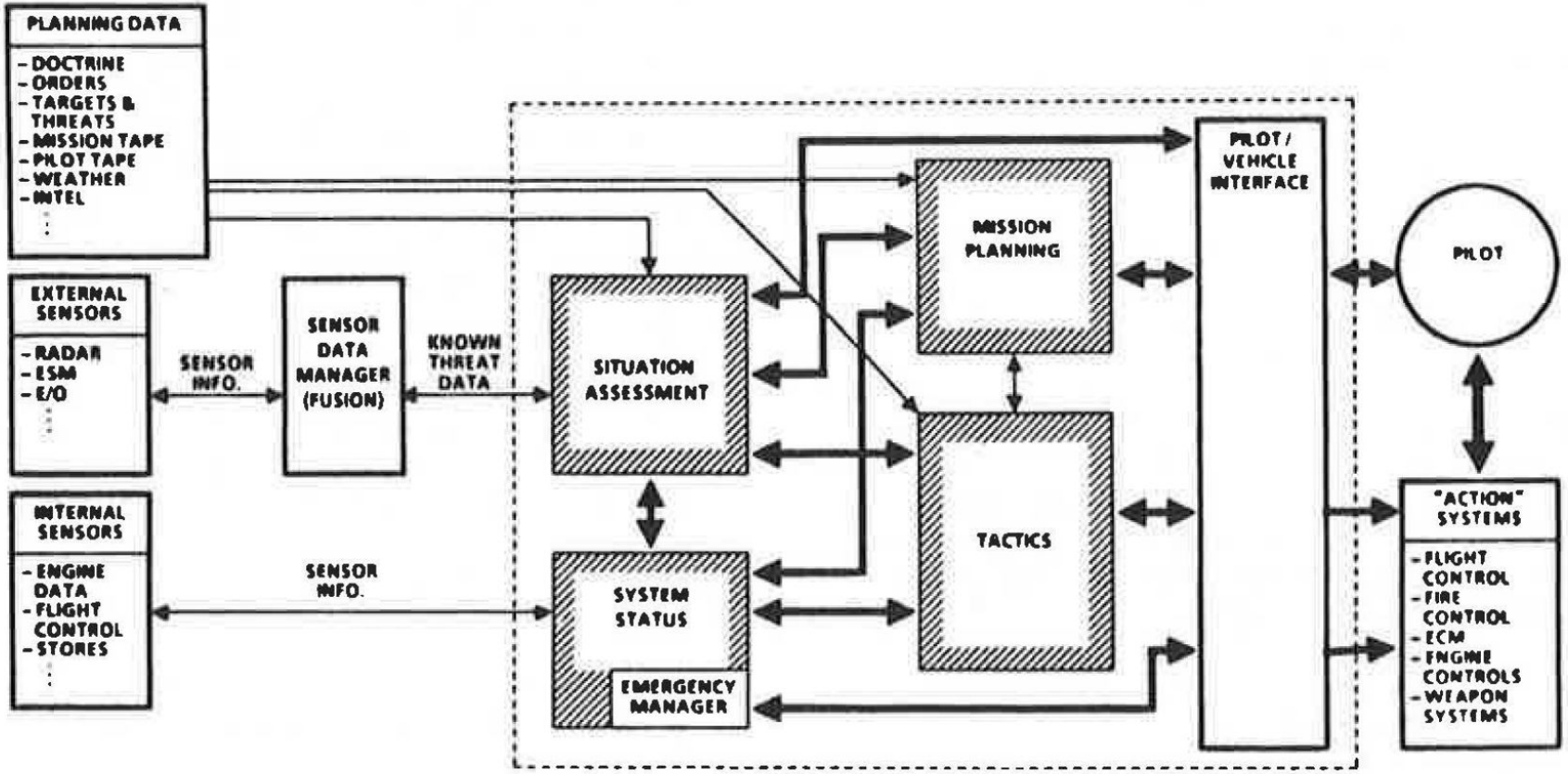


Figure 1: Terregator


ALV
WARNING
ALV

Project 2) Pilot's Associate

- Intelligent fighter associate for the Air Force
- Automated copilot
- Keeping track of many functions pilots must monitor
- It could take over some of the pilot's tasks, reroute electrical signals/fluids in the event of damage.
- Voice recognition was a primary goal → the assistant would be able to understand spoken English even amidst the noise of the jet cockpit.



PILOT'S ASSOCIATE

Figure 6. Pilot's Associate Mission.

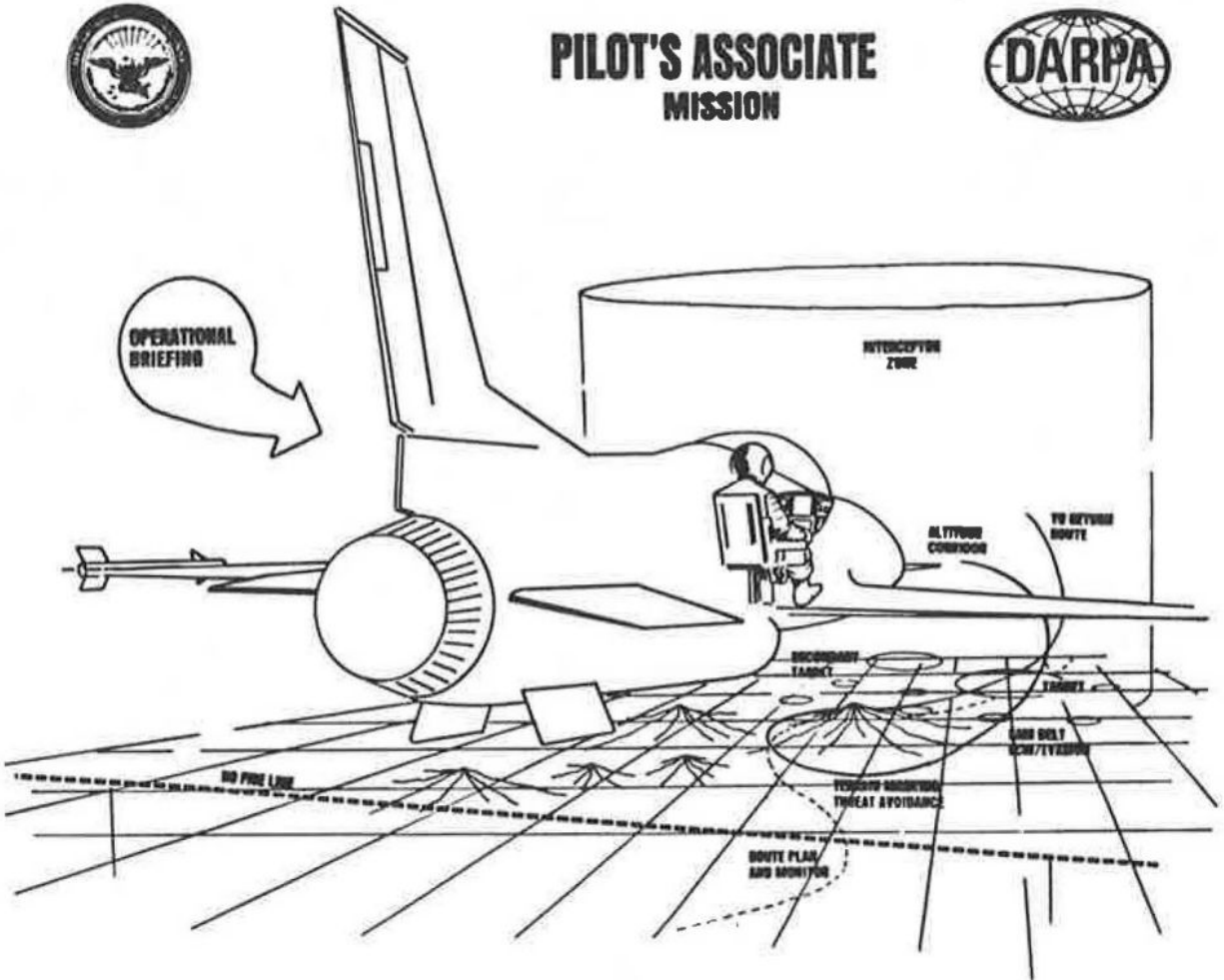
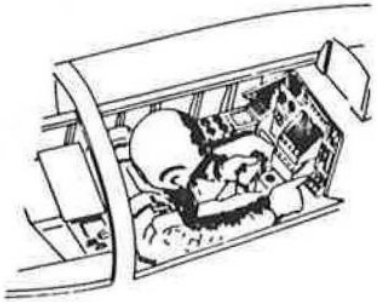


Figure 3. Pilot's Associate Program Overview.



PROGRAM OBJECTIVES

- EXPLOIT DARPA STRATEGIC COMPUTING TECHNOLOGY BASE
  - EXPERT SYSTEMS
  - PARALLEL PROCESSING
  - SPEECH SYSTEMS
  - PILOT-VEHICLE INTERFACE
- REAL-TIME INFORMATION MANAGEMENT
- PROVIDE DECISION AIDING IN DEMANDING HIGH THREAT SCENARIOS

FUNCTIONAL AREAS

- SITUATION ASSESSMENT
- SYSTEM STATUS
- MISSION PLANNING
- TACTICS ADVISORY

DEVELOPMENT ENVIRONMENT

- COMPUTER LABORATORY — PART TASK SIMULATORS
- FULL MISSION SIMULATORS

8643/8SW/GW28/1

PROGRAM MILESTONES

FY86-88:	PHASE I — KNOWLEDGE ENGINEERING, SYSTEM DEVELOPMENT
FY86:	DEMO ONE — FEASIBILITY AND IMPACT ASSESSMENT
FY87:	DEMO TWO — NON-REAL-TIME PART TASK SYSTEM SIMULATION
FY88:	DEMO THREE — NEAR-REAL-TIME TOTAL SYSTEM SIMULATION
FY89-90:	PHASE II — INTEGRATION OF TECHNOLOGY BASE PRODUCTS
FY90:	DEMO FOUR — FULL MISSION REAL-TIME SIMULATION, TOTAL SYSTEM EVALUATION

Project 3) Battle management expert system

- battle management for aircraft carrier battle groups
- analyze sensor data from the carrier's huge field of engagement, interpret threats, keep track of friendly forces, and plan response options that it would then "suggest" to the battle group commanders
- DARPA later (in 1985) added another battle manager for the Army's AirLand Battle Doctrine plan for forward air and ground combat in Europe



- What is DARPA’s legacy?
- What DARPA tech do you use today?
- What are the pros and cons of government funded tech research?

Autonomous Weapons

“Any weapons system with autonomy in its critical functions. That is, a weapon system which can select (i.e. search for or detect, identify, track, select) and attack (i.e. use force against, neutralize, damage or destroy) target without human intervention.”
— International Committee of the Red Cross

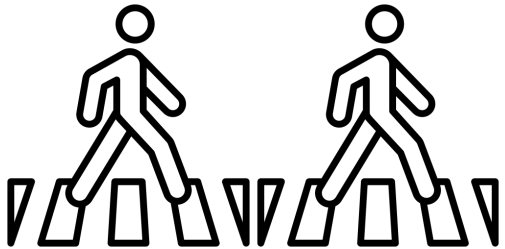


Trolley Problem for Machines

Autonomous car has to decide colliding with single pedestrian vs. multiple pedestrians or to save the car's passenger.







When autonomous machines make decisions about life-and-death situations

When it comes to how agents SHOULD decide: people tend to apply similar norms to human and artificial decision makers

But when people judge the moral quality of the agents' actual decision, their judgements tend to differ:

1. People imagine the the decision conflict the human agent endures → blame mitigation
2. Harder to imagine the machine decision maker's processes and justifications → higher blame

→ Even if the ex ante norms & decisions are the same, machines are judged differently for their actions

Scheutz & Malle. "May Machines Take Lives to Save Lives?" Lethal Autonomous Weapons. Galliott et al.

- What happens are machines are equipped to make life-death decisions?
- What “norms” should they optimize?
- Should they be equipped to justify their “choices”?